RDL Tool users manual

RSO-14-005355-06

2016/1/25 Rev. 0.60

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Introduction

Introduction

This application is used to generate Display Lists (hereafter referred to as DL) to be used by the VOWE driver for RH850/D1M. Please use it as technical reference for the software development.

This application is for reference, we do not guarantee its operation. If you want to include in the actual system, after thoroughly evaluation of the whole system, please final decision on the applicability of the information of customer.

Introduction

This application can be as follows.

[RH850/D1M]

Create a DL header in C Programming Language for the VOWE driver. Or preview with DL intermediate file format (text file).

Execution environment

Execution environment

Execution environment such as the following is required to run this application.

OS	• Windows7 (Even Windows Vista, will probably work)
Library	• .NET Framework version 4.5
Monitor	• XGA or greater

If above library is not installed on your computer, please download it from the Microsoft website linked below.

.NET Framework : http://www.microsoft.com/**en-us**/download/details.aspx?id=30653

(The part that is written in red, may vary depending on your language.)

Execution environment

The application provides an installation / uninstallation routine as described below.

[Install]

Execute the installation program setup_*RDL_Tool_x.x.x.x_xxxxxxxx.exe.*

[Uninstall]

The uninstallation can be started by clicking the Start Menu entry:

Renesas Electronics > RDL Tool > Uninstall

File Structures

File Structures

File structure of this application is as follows.

- RDL_Tool_Ver.x.x.x.x_xxxxxxxx (Directory)
  - RDL_Tool.exe (File)
  - Inter-ProcessCommunication.dll (File)
  - Filters (Directory)
    - fihseye_lite.exe (File)
    - FanwiseDL.exe (File)
    - TextToDL.exe (File)
  - operationTools (Directory)
    - DLGenerator.exe (File)
    - vowe_render.exe (File)
    - voweemu.dll (File)
    - Inter-ProcessCommunication.dll (File)

Procedure of DL generation

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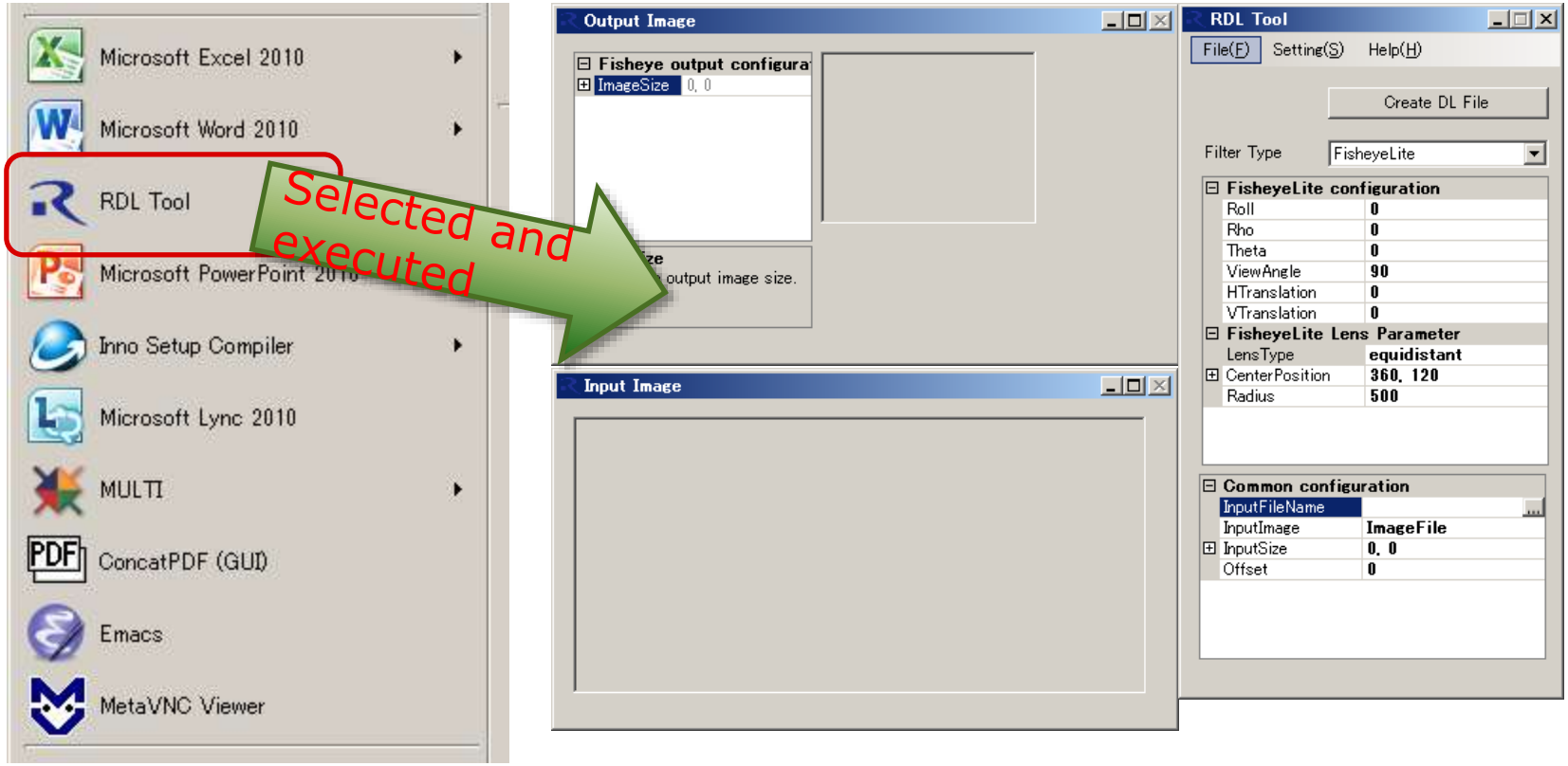
Procedure of DL generation

Step1.

To select and execute "RDL Tool" from the Windows Start menu.



Procedure of DL generation

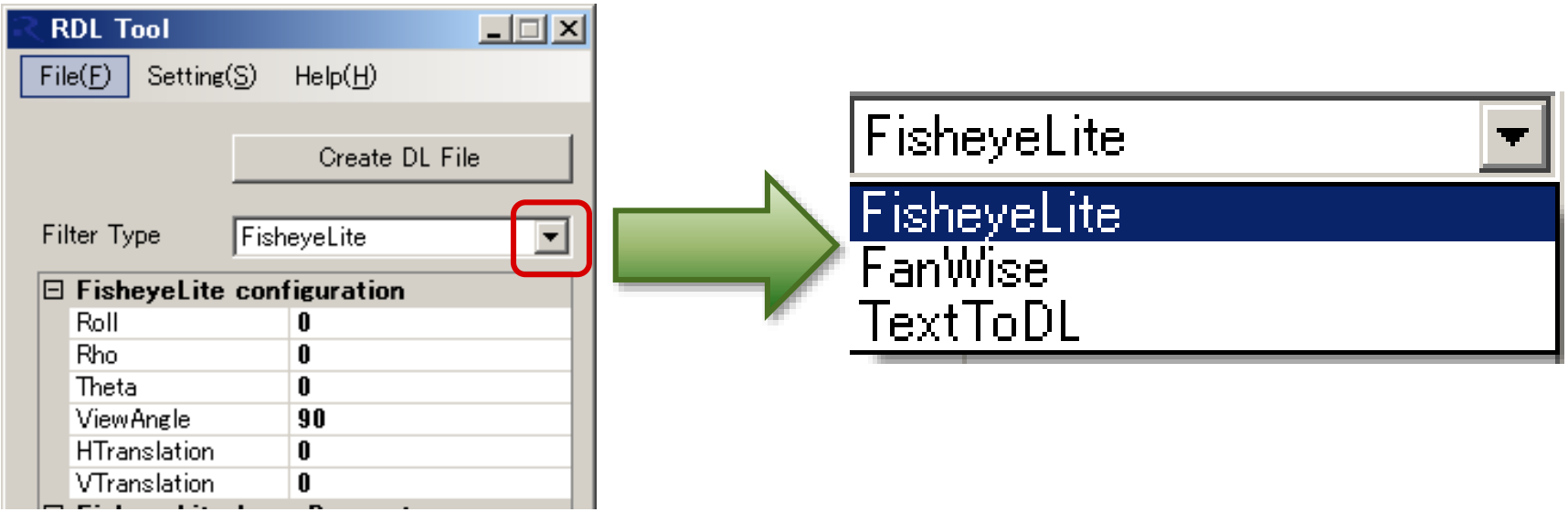
Step2.

Start with the main window of the RDL Tool as shown below and select “Filter Type”. (The Tool remembers the settings you made in your last session.) Selecting a Filter gives you various parameters specific to that filter in the window below. There are three kinds of choices: "TextToDL", "FisheyeLite", and "Fanwise".

“FisheyeLite” is a fish-eye warping parameter specification.

“FanWise” is a fan-shapes and trapezoid deformation parameter specification.

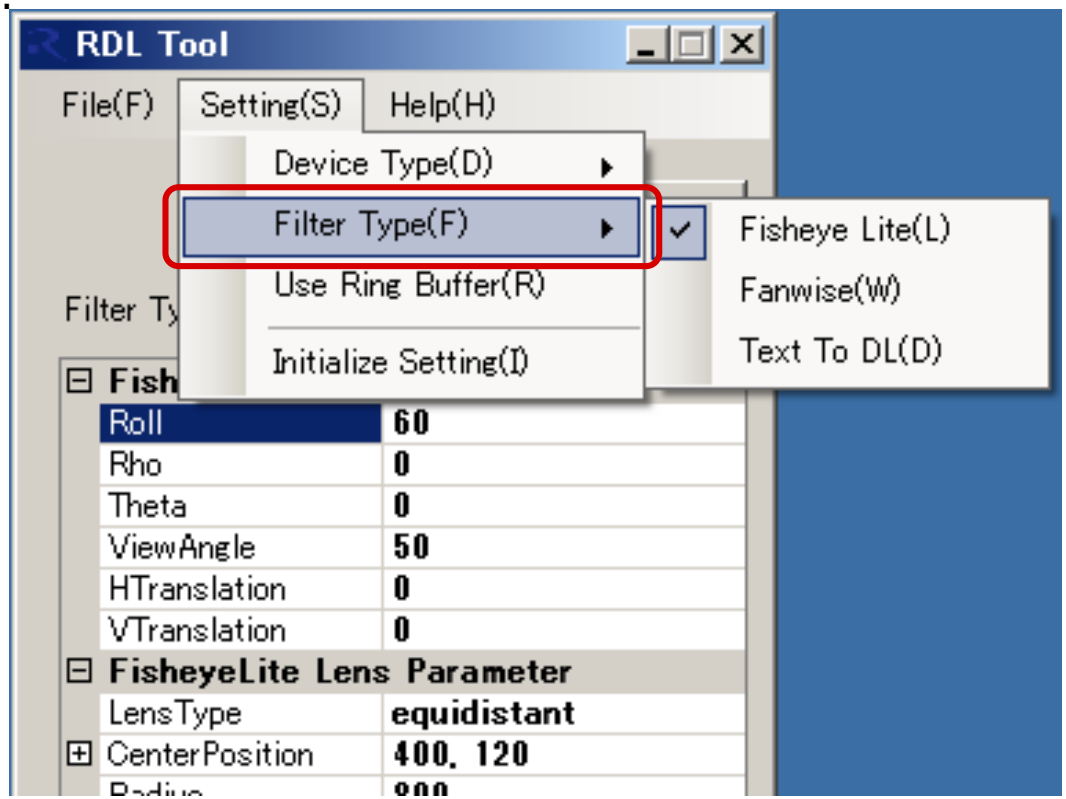
“Text file to DL” is a filter to create DL from custom coordinate data.



Procedure of DL generation

Step2.1.

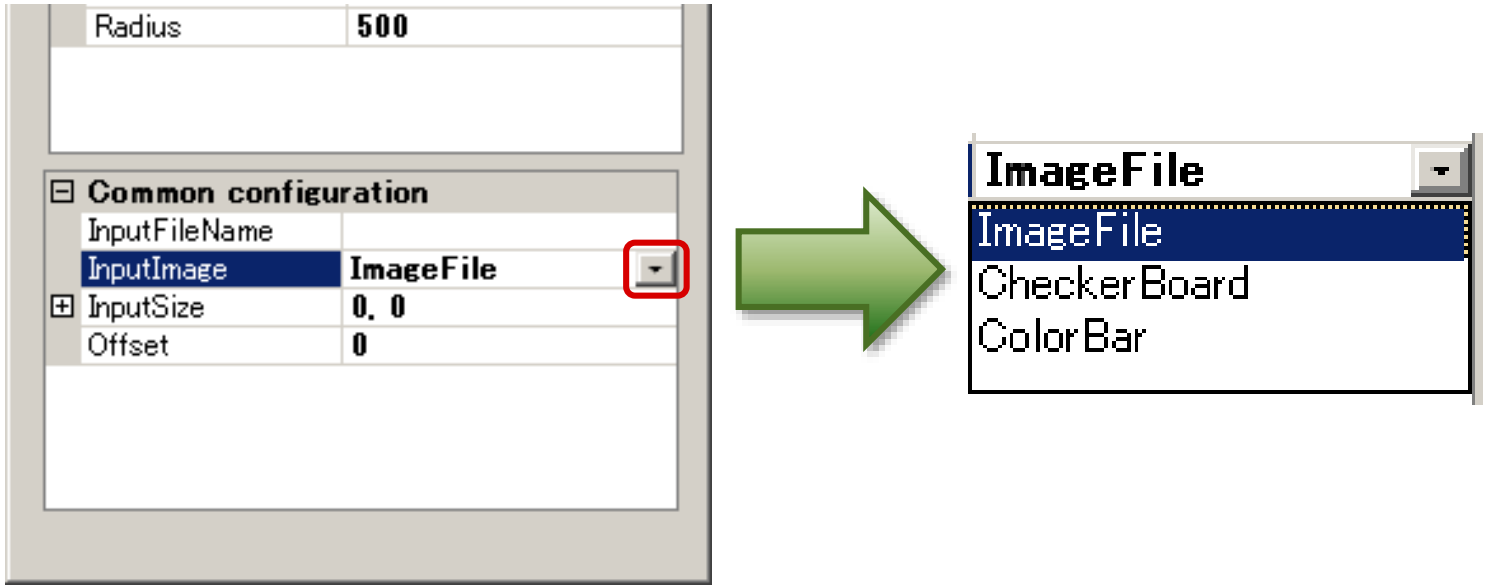
The choice of filter type can also be done by selecting "Filter Type(F)" from the "Settings(S)" menu.



Procedure of DL generation

Step3.

Select the type of the preview image used to show the effect of the selected warping function. If you select "ImageFile" for the setting "InputImage" in the Common configuration area, continue to Step 3.1 to provide a custom image file. For the other options, skip the next Step.

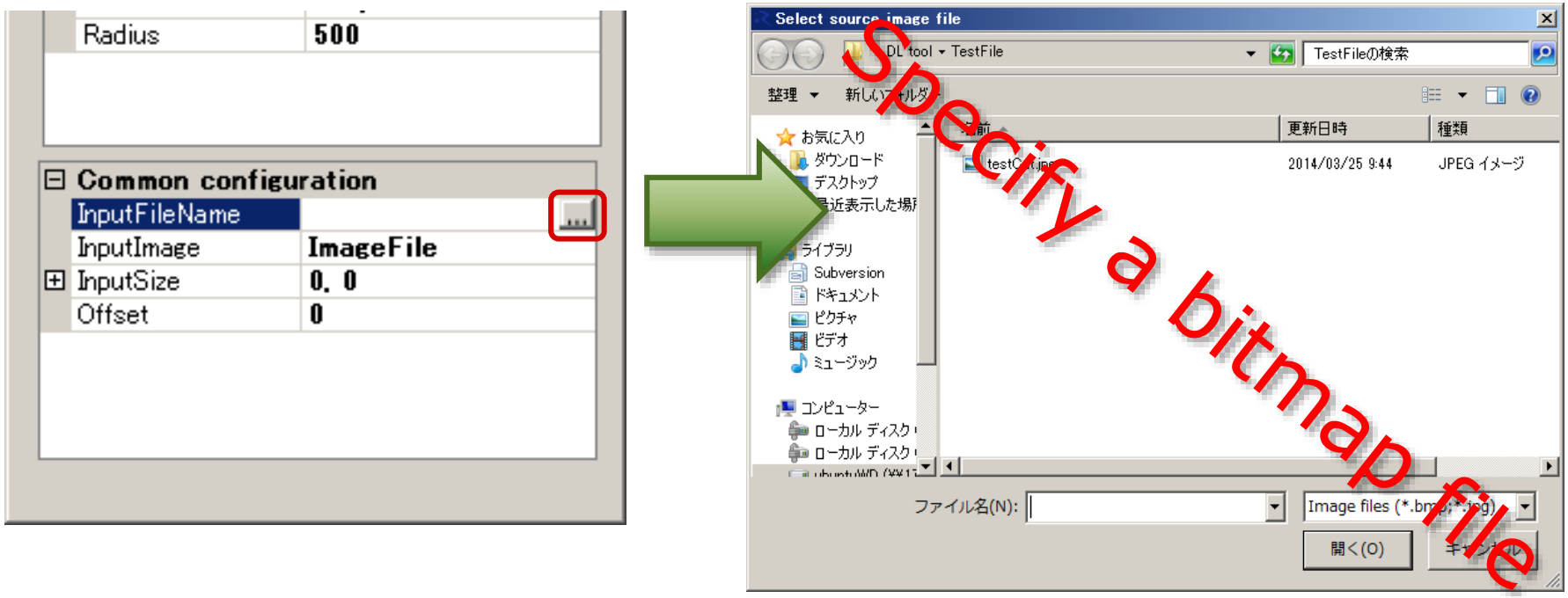


Procedure of DL generation

Step3.1. (If you select “ImageFile” at Step3.)

Specify a bitmap file as the source image in “InputFileName” edit box of “Common configuration” property sheet.

This parameter can be specified in two ways. You may use the "Open" dialog by pressing the "..." button on the right side of the input field. Alternatively you may type directly into the input field. It can be an absolute or a relative path. If a relative path specified, it is relative designation from “DL_Tool.exe”.



Procedure of DL generation

Step3.2. (If you select “ImageFile” at Step3.)

If you specify a source image, the source image is displayed in the “Input Image” window. In addition, after it has been warped using the current parameters will be displayed in the “Output Image” window.

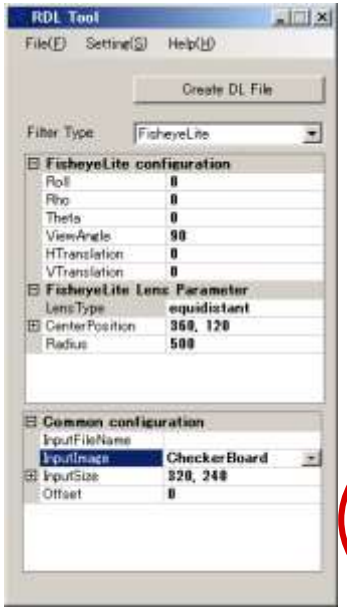
Input Image

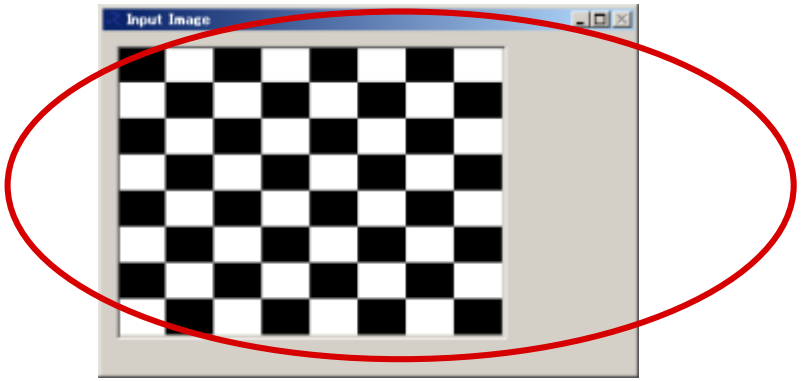
Output Image

Procedure of DL generation

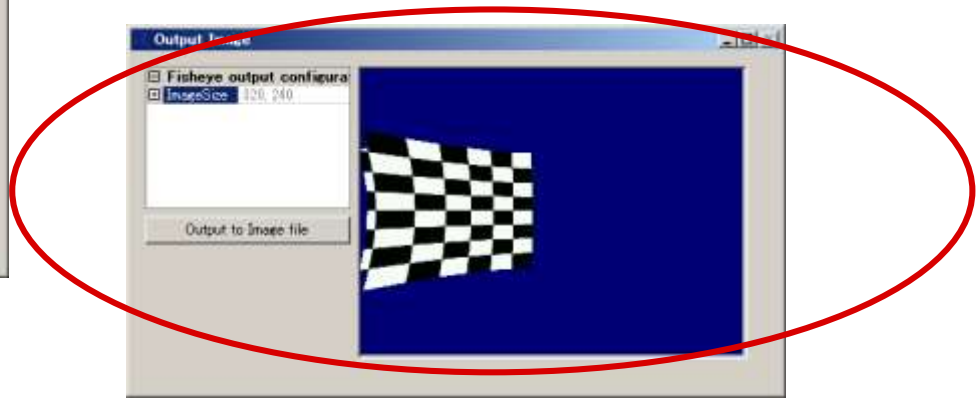
Step3.3. (If you select “CheckerBoard” or “ColorBar” at Step3.)

If you select the "color bar" or "checkerboard", an automatically generated image is displayed in the "input image" window. The resolution specified in "InputSize" in the Common Configuration area is used for the automatically generated image as well as for the output image. It is automatically set to “320, 240" if any value of the "InputSize" column is 0.





Input Image

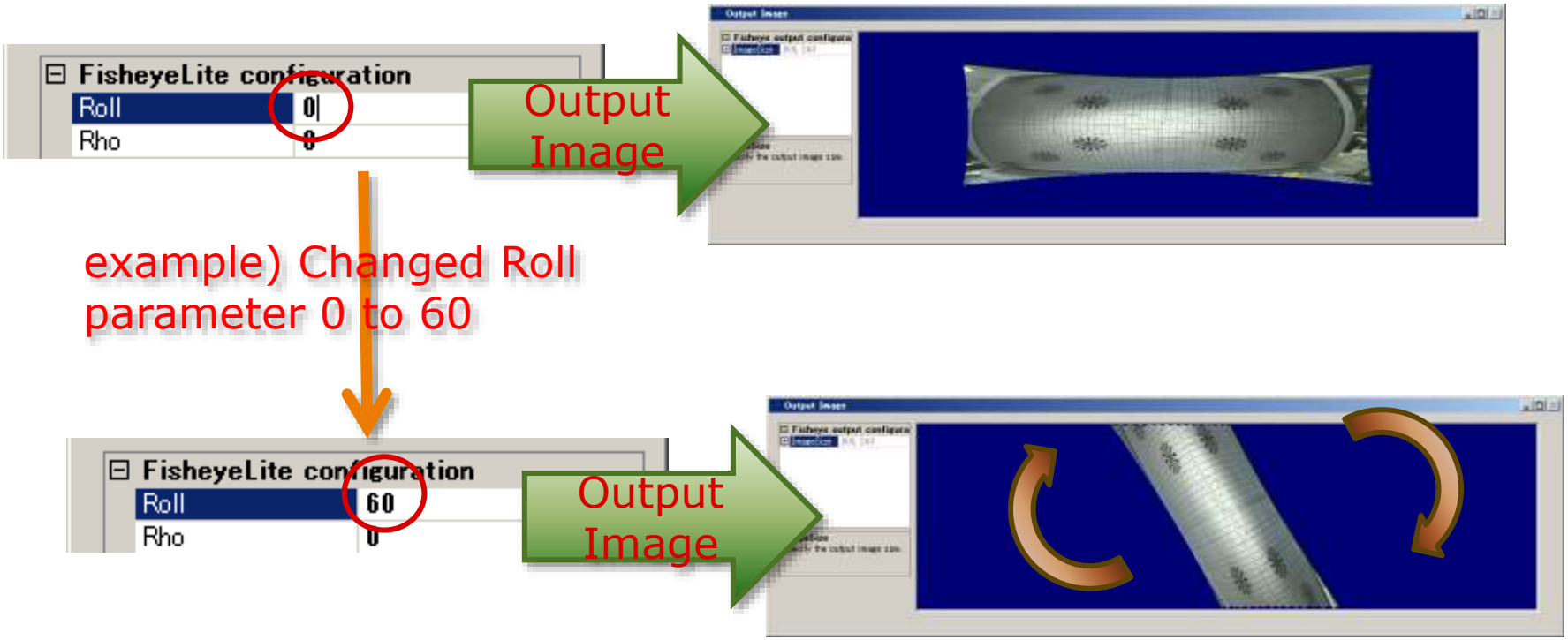


Output Image

Procedure of DL generation

Step4.

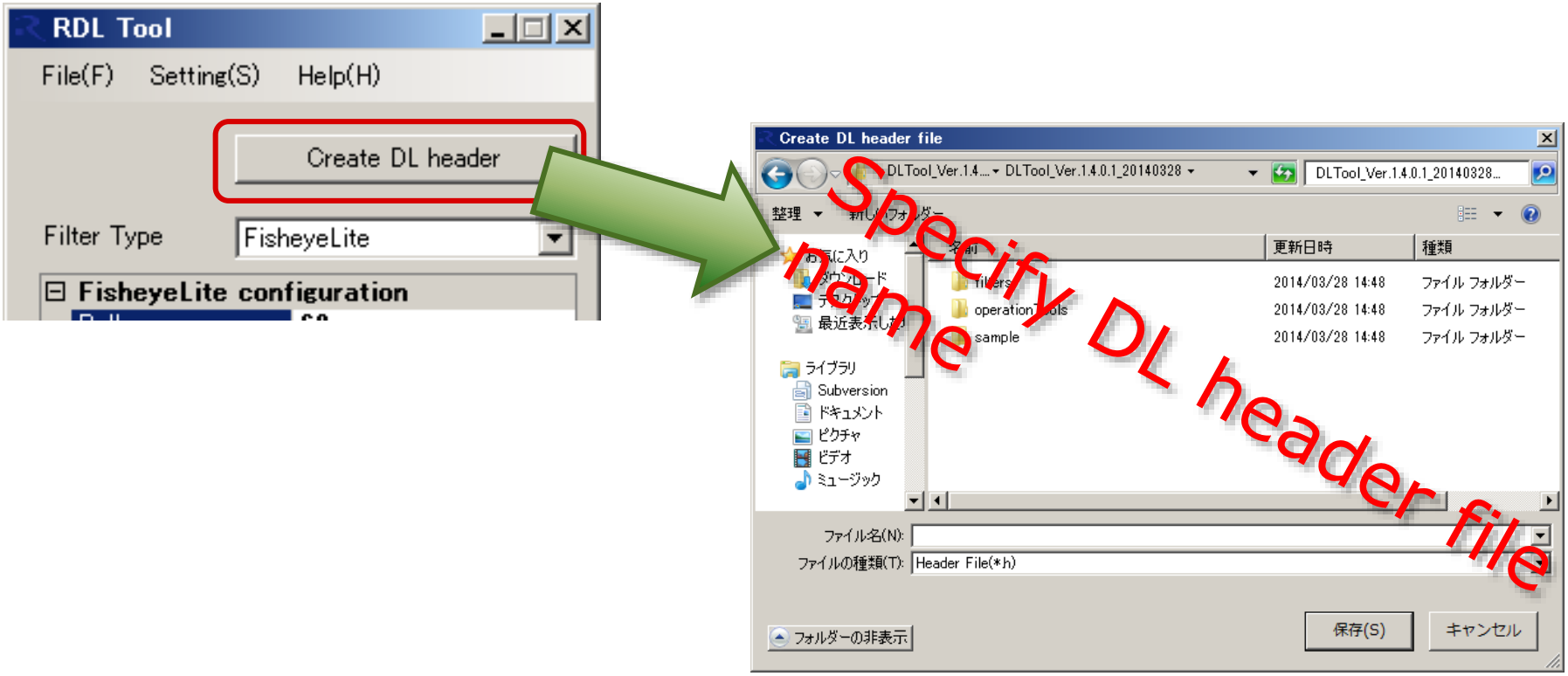
Use the Filter configuration area to configure the filter parameters. (Please refer to chapter of "Parameter Description" for more information about the parameters). You will see the warped image is automatically reflected in the "Output Image" window when you set parameters.



Procedure of DL generation

Step5.

Press the “Create DL Header” button to create the DL header file for the C language.



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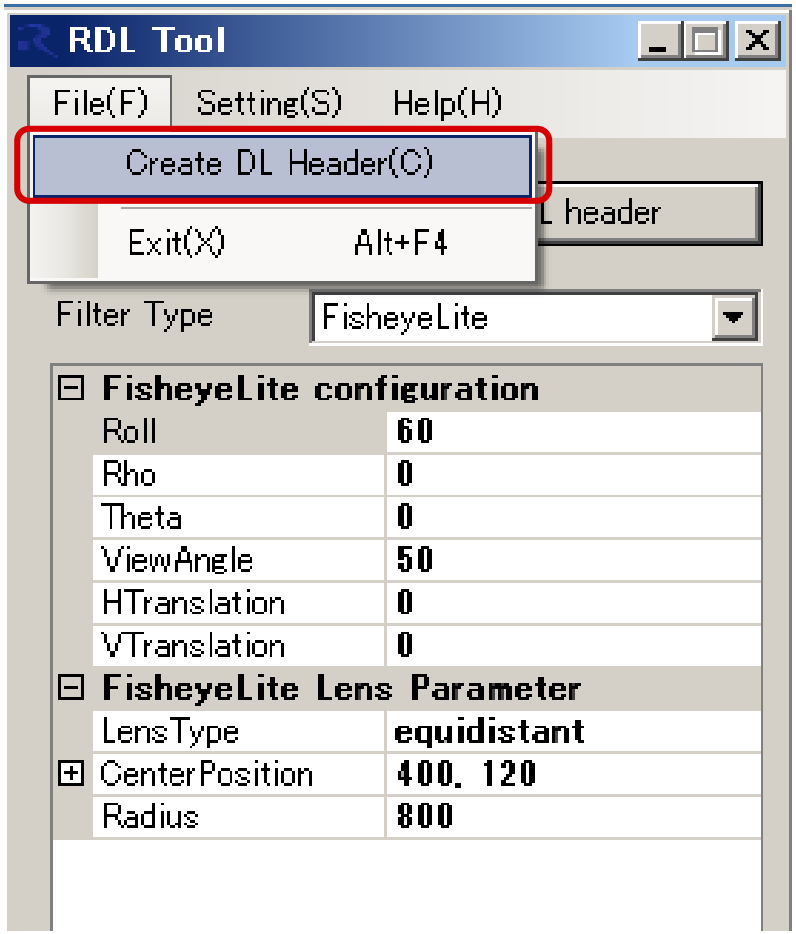
Procedure of DL generation

Step5.1.

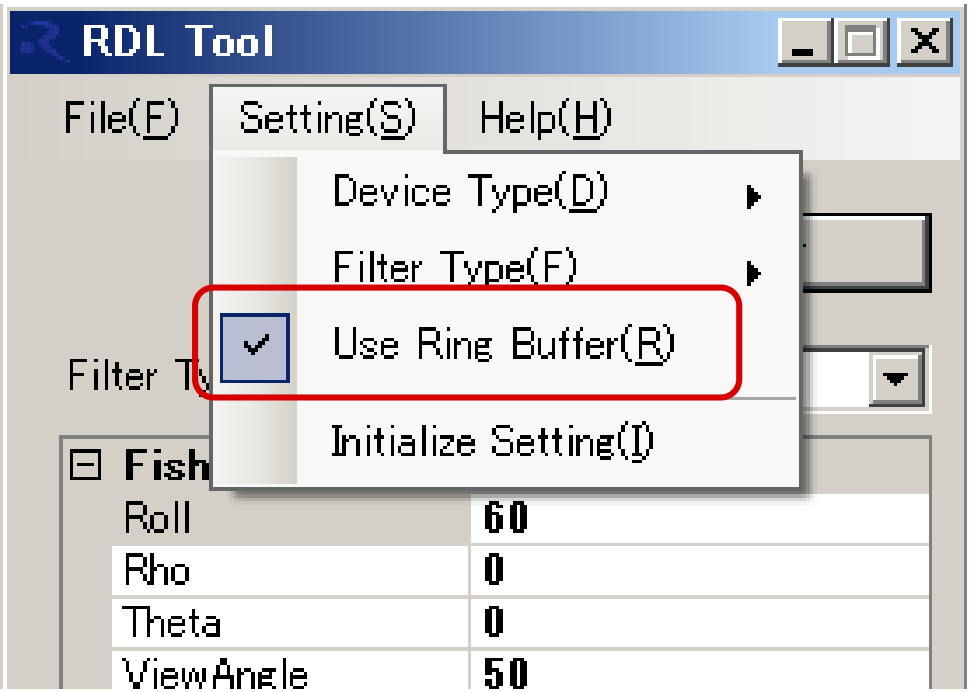
Alternatively, select "Create DL Header(C)" from the "File(F)" menu to create a header file.



Procedure of DL generation

Step5.2.

If you are using the ring buffer mode of the VOWE, you need to create a dedicated DL. For this purpose, please activate the entry "Use Ring Buffer(R)" from the menu "Settings(S)". You can create DL of the ring buffer mode only when you run the "Create DL header" in that state.



Parameter Description

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Parameter Description

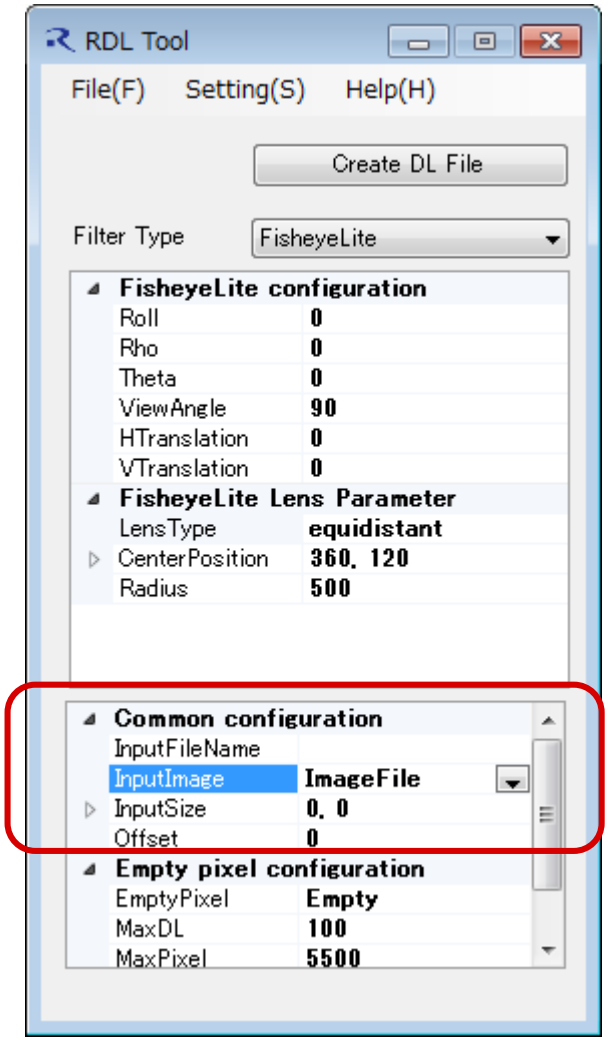
Parameters can be classified into the following categories:

- **Common configuration**
- **FisheyeLite” configuration**
- **Fanwise configuration**
- **Text file to DL” configuration**
- **Empty pixel configuration**

The common configuration is always visible.

The other categories are only visible, if their settings are applicable to the currently selected filter.

About Common configuration. (1)



"Common configuration" will be displayed in this area at any time.

About Common configuration. (2)

InputImage

- Specify from following types of input image.
  - ImageFile
  - CheckerBoard
  - ColorBar

InputFileName

- Specify the input image file, if “ImageFile” is selected as input image.

InputSize

- Specify the input image size. When the input image file is specified, the size of the specified image will be set automatically, but can be changed manually.

Offset

- Specify an offset as a multiple of 32. This offset makes it possible to use negative coordinates in top (-Y) and left (-X) direction. This way triangles may intersect top and left screen borders.

About Common configuration. (3)

InputImage

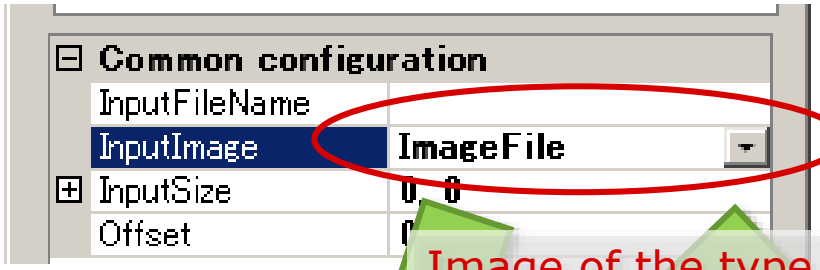
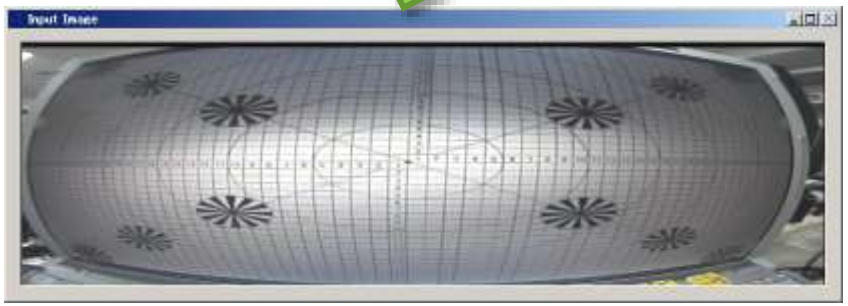


Image of the type specified is displayed on Input Image window.

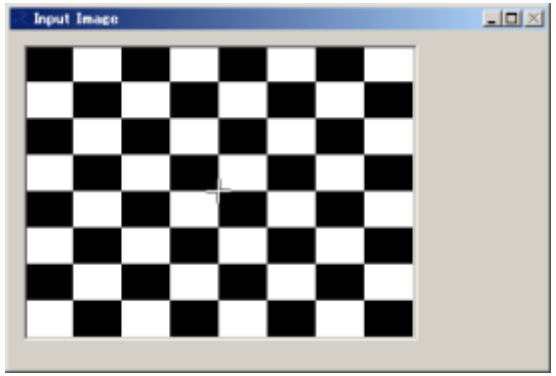
ColorBar

ImageFile



ImageFile: Use specified image file.

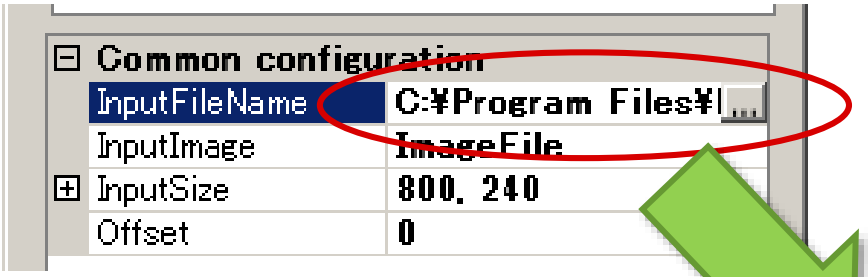
CheckerBoard



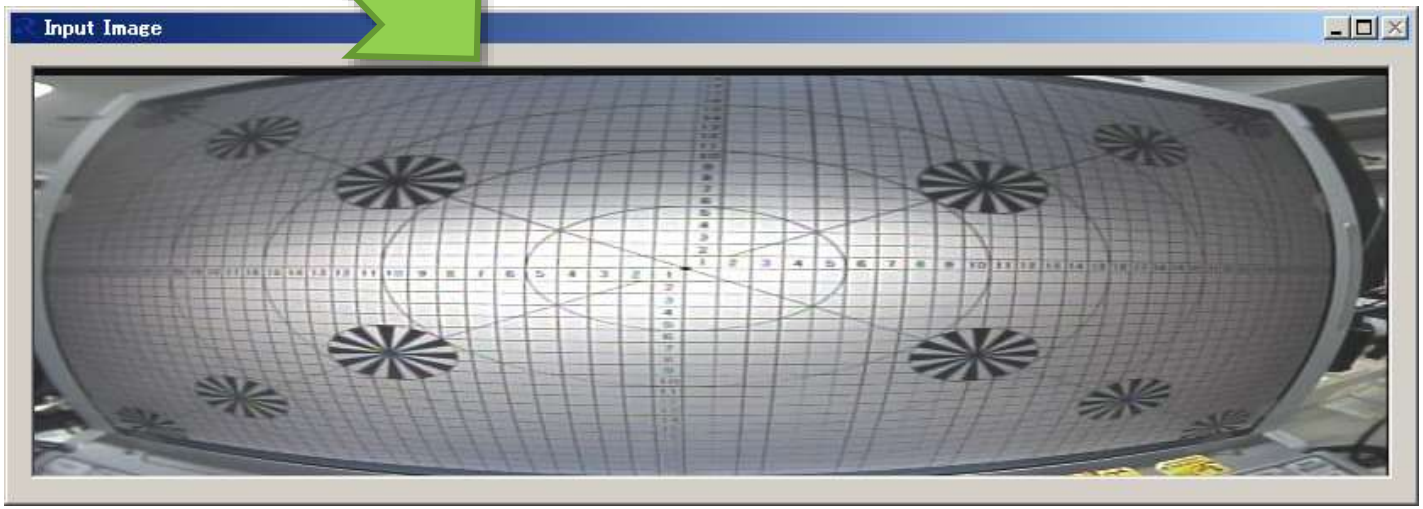
ColorBar / CheckerBoard:
Use preset image.

About Common configuration. (4)

InputFileName

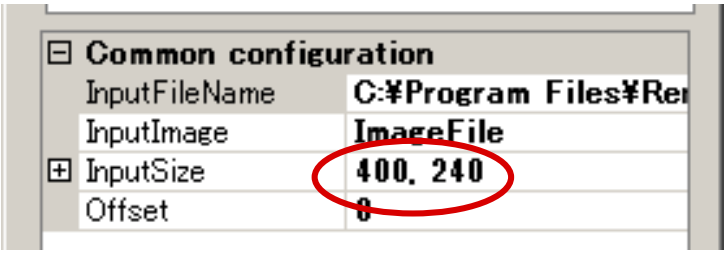


Specified bitmap file is displayed on Input Image window.

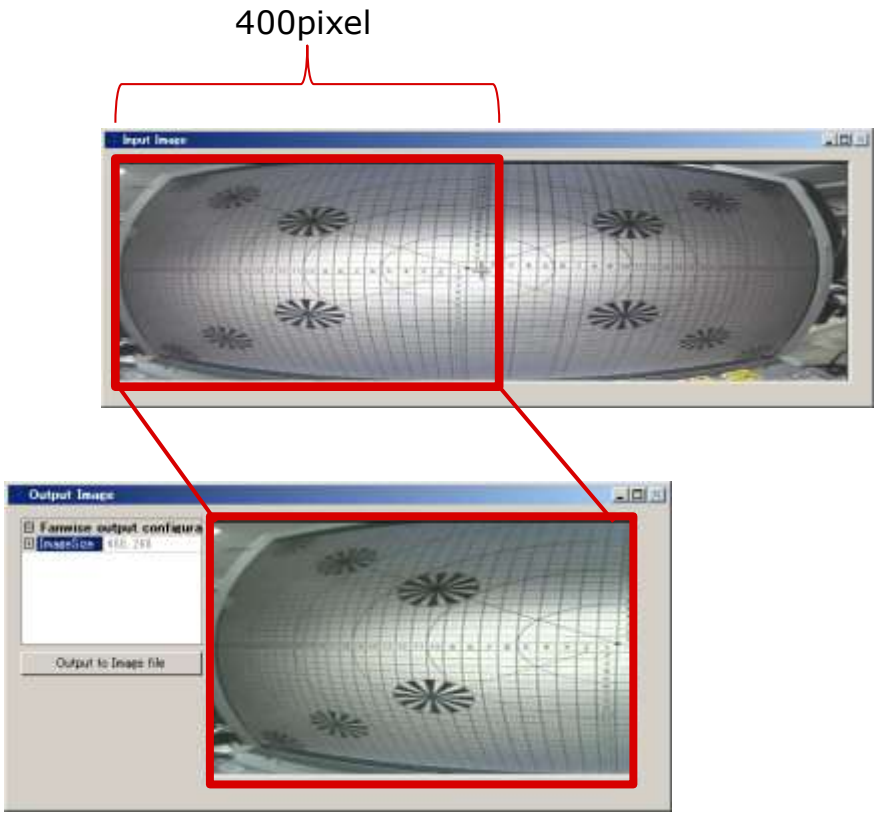


About Common configuration. (5)

InputSize (1) - When selecting "ImageFile" in "InputImage"

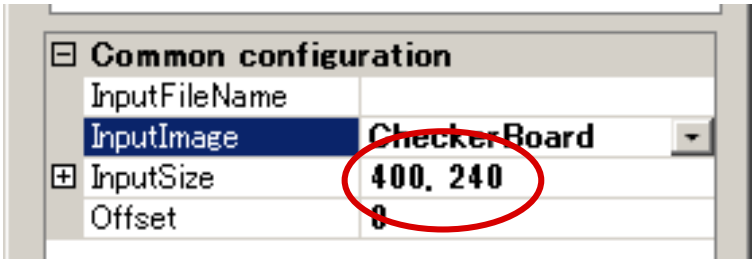


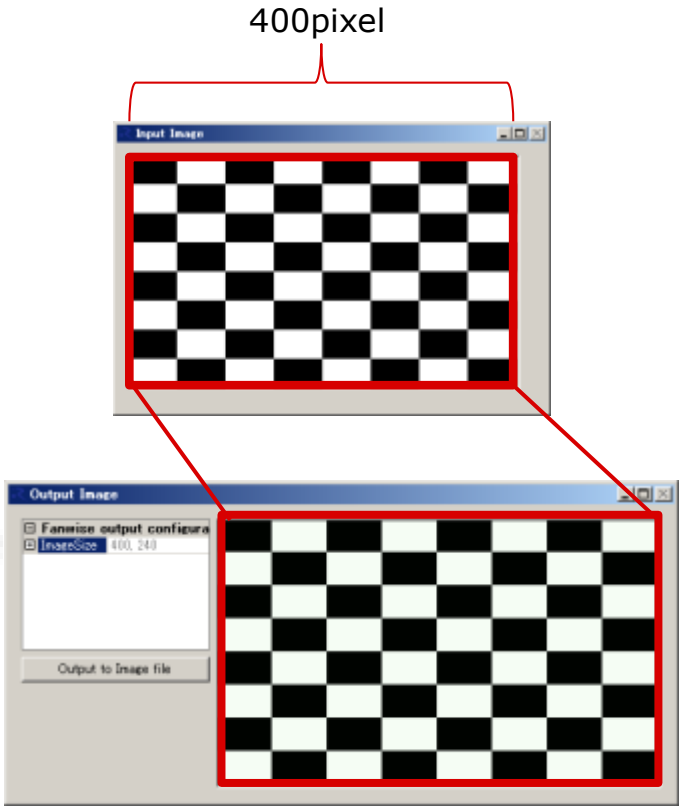
If you manually changed the InputSize, only specify the size of the input image is warped.



About Common configuration. (6)

InputSize (2) - When selecting "ColorBar" or "CheckerBoard" in "InputImage"





If you change the InputSize, output image size and input image size is changed to match it.

About Common configuration. (7)

Offset

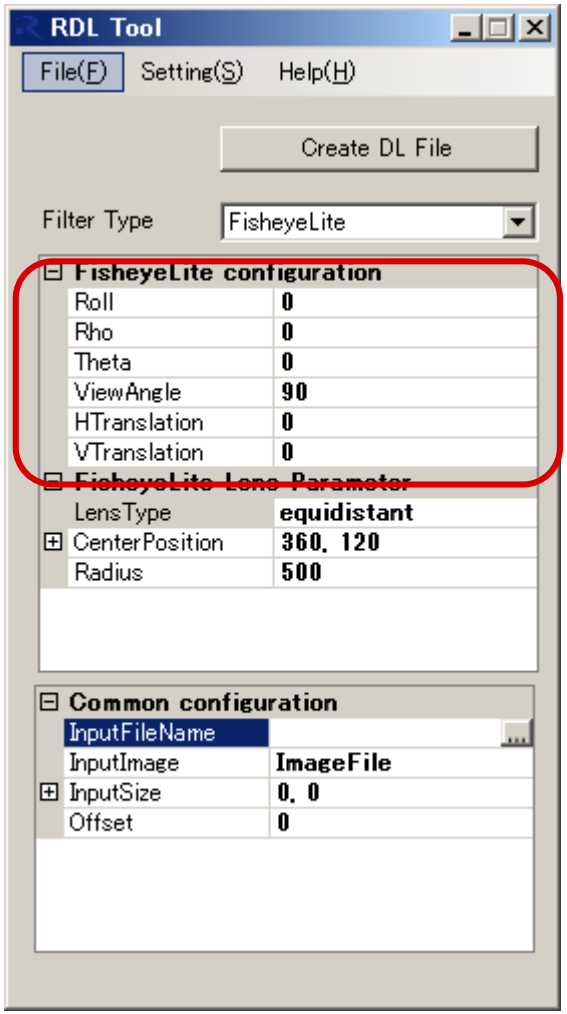
Offset = 0

Offset = 32

Negative coordinates are not supported by the warping hardware. Triangles with negative coordinates are removed, this may result in the staircase effect as shown on left side. With this function the origin of the coordinate system is moved by "Offset". Negative coordinates of up to "-Offset" are now possible.
Due to hardware constraints, Offset must be a multiple of 32.

If you are using the ring buffer mode of the VOWE, you should set “Offset = 0”.
If you are using the frame buffer mode with “Offset > 0”, stride for “Offset” pixels is needed in VOWE buffer. (Stride >= Width + Offset)

About FisheyeLite configuration. (1)



"FisheyeLite configuration" Will be displayed in this area when "FisheyeLite" is selected to "Filter Type".

About FisheyeLite configuration. (2)

Roll (-180 ~ +180)

- Specifies the rotation angle on the screen plane.

Rho (-90 ~ +90)

- Specify the swing angle in vertical direction.(around horizontal axis)

Theta (-90 ~ +90)

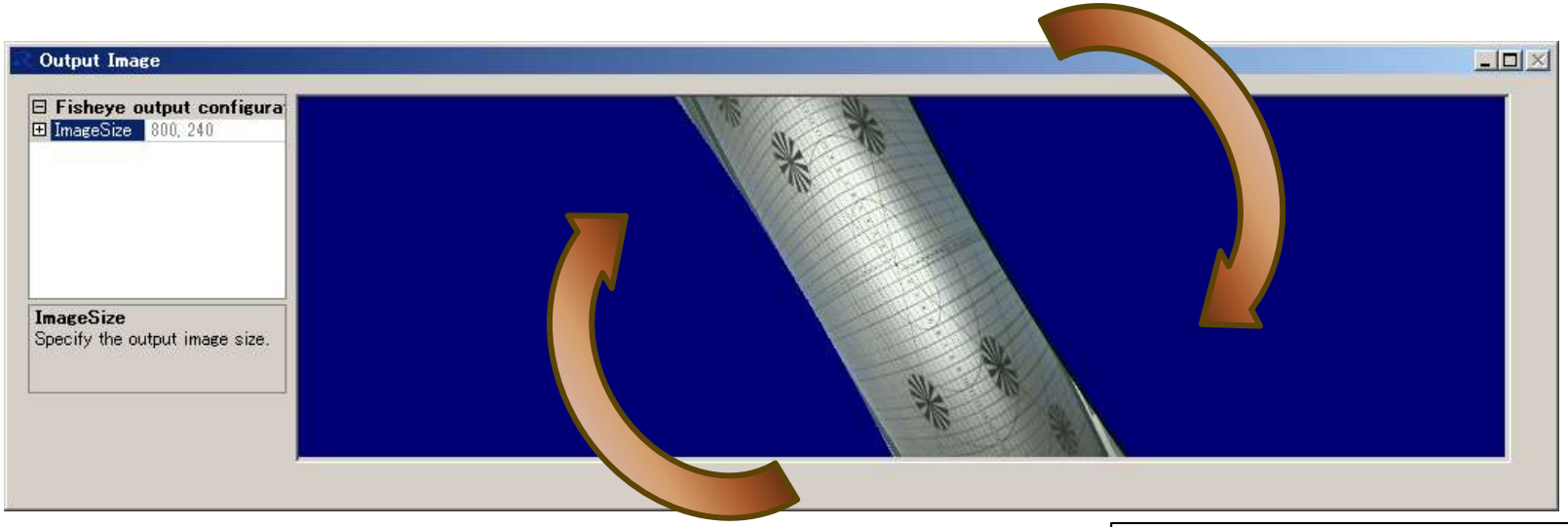
- Specify the swing angle in horizontal direction. (around vertical axis)

ViewAngle (-170 ~ +170)

- Specify the vertical viewing angle. A wider viewing angle results in a smaller picture.

About FisheyeLite configuration. (3)

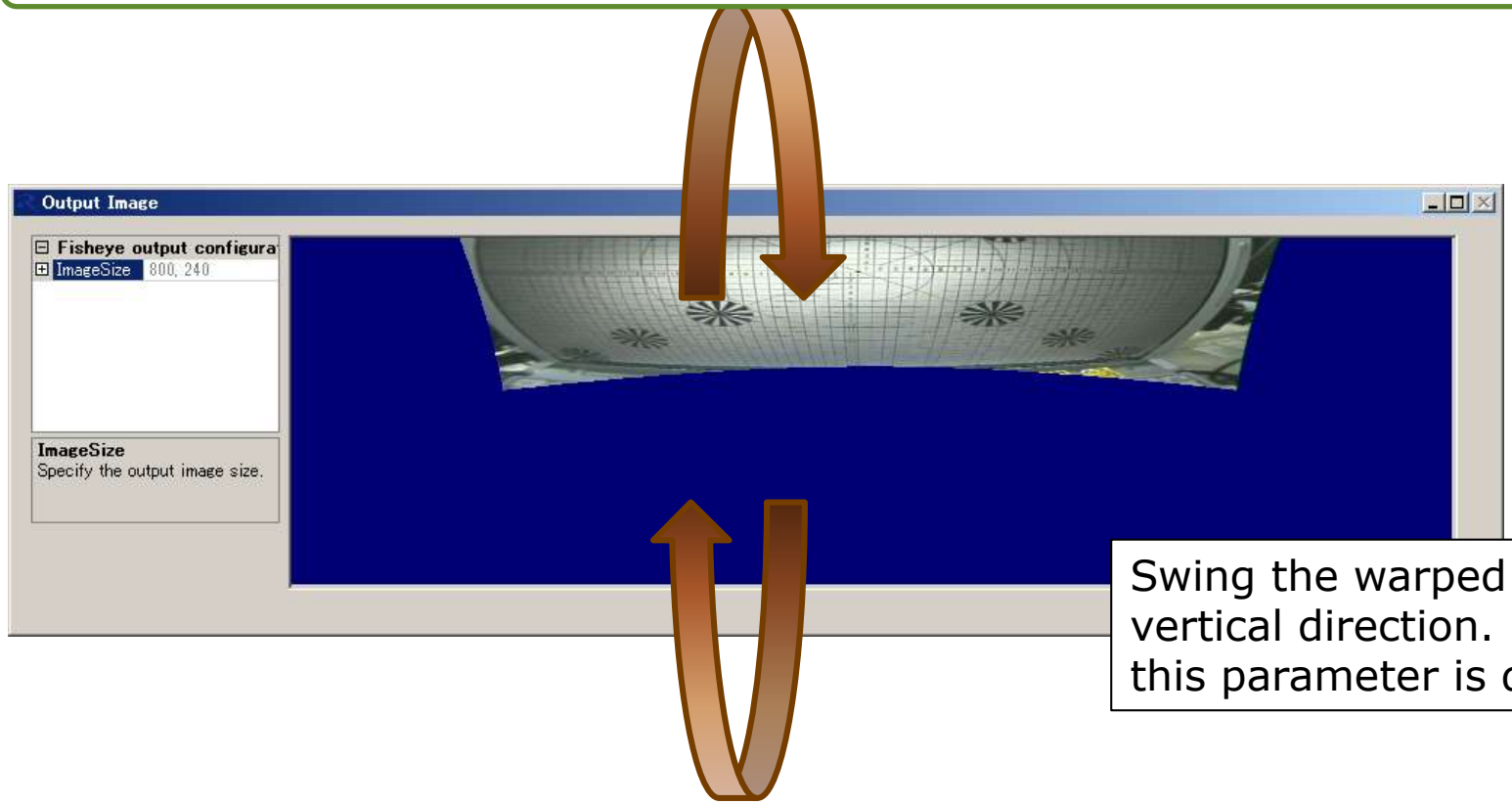
Roll



Rotate the warped image. Unit of this parameter is degree.

About FisheyeLite configuration. (4)

Rho



Swing the warped image in vertical direction. Unit of this parameter is degree.

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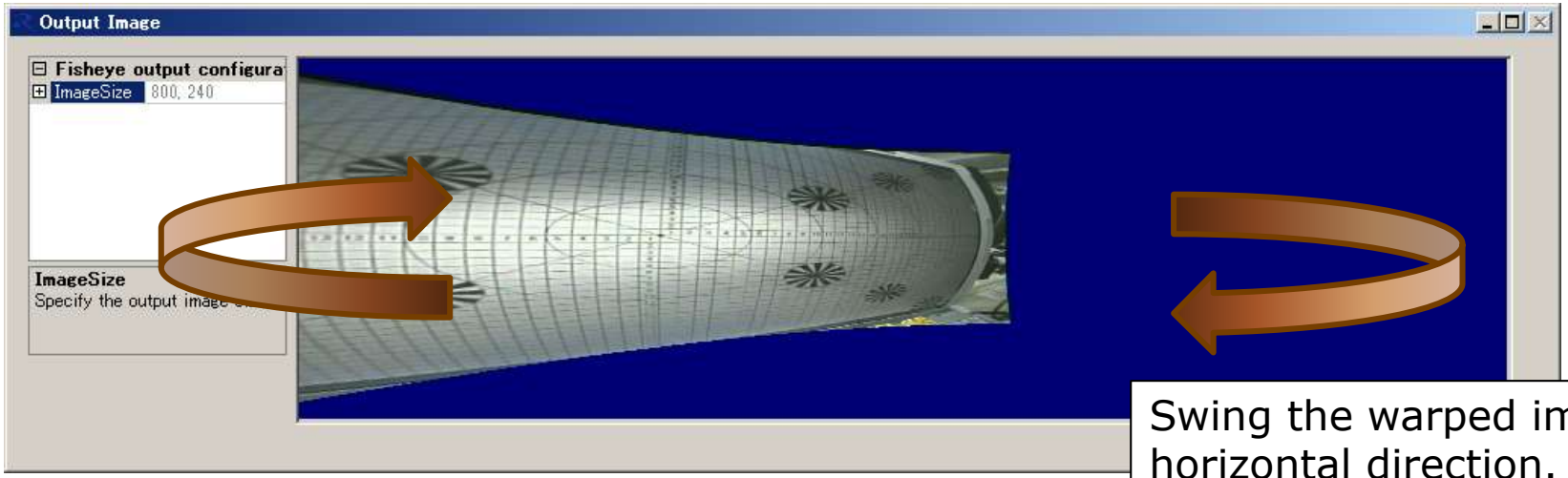
About FisheyeLite configuration. (5)

Theta



Swing the warped image in horizontal direction. Unit of this parameter is degree.

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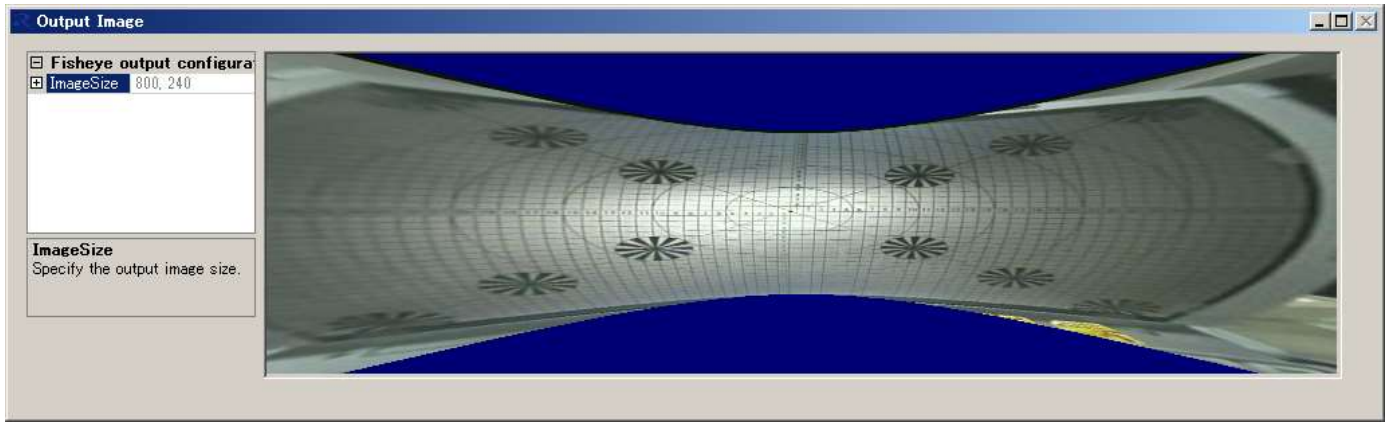
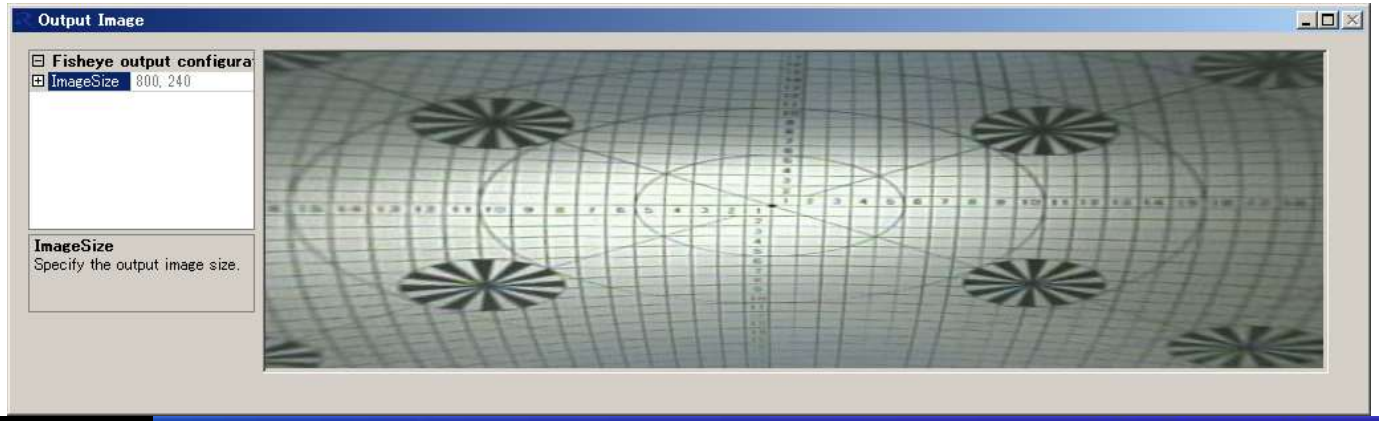
About FisheyeLite configuration. (6)

ViewAngle

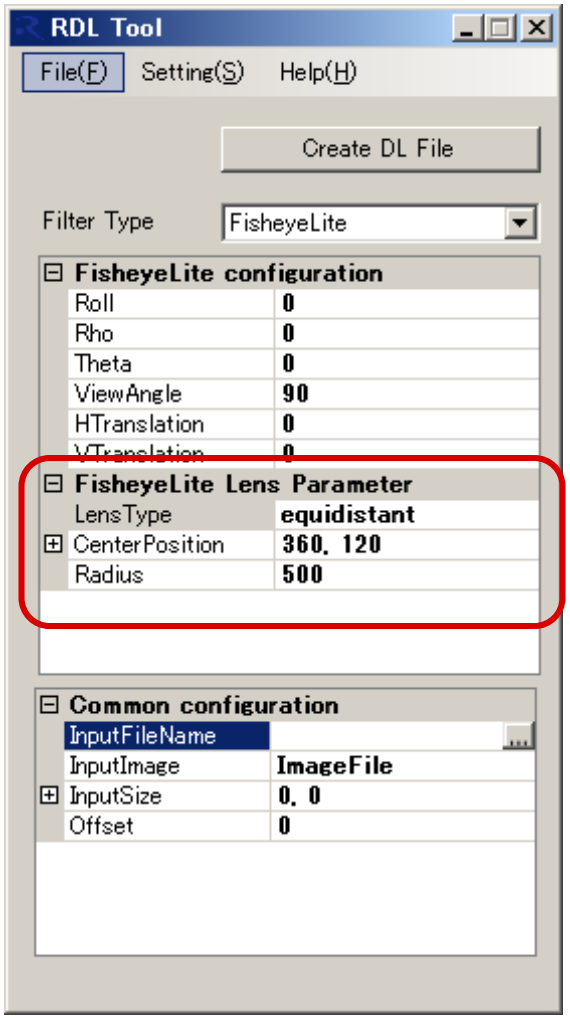
ViewAngle = 90



ViewAngle = 50



About FisheyeLite configuration. (7)



"FisheyeLite Lens Parameter"
Will be displayed in this area when "FisheyeLite" is selected to "Filter Type".

About FisheyeLite configuration. (8)

CenterPosition

- Specify the center coordinates of the distortion of the input image.

LensType

- Specify the type of fish-eye lens. Algorithm of warping will change.
  - *Equidistant* - equidistant projection
  - *Stereographic* - stereographic projection

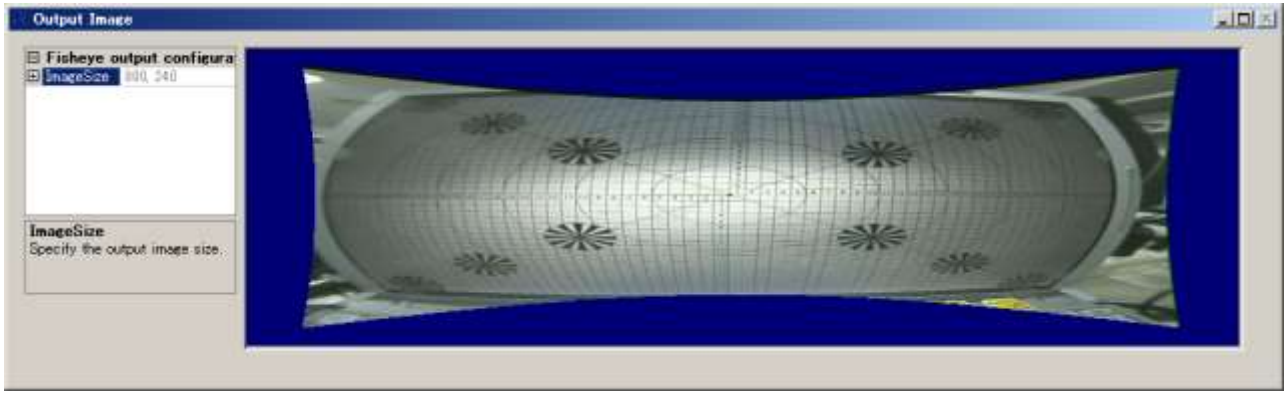
Radius

- Specify the radius of the image distortion. The smaller the value, the warp amount of distortion increases.

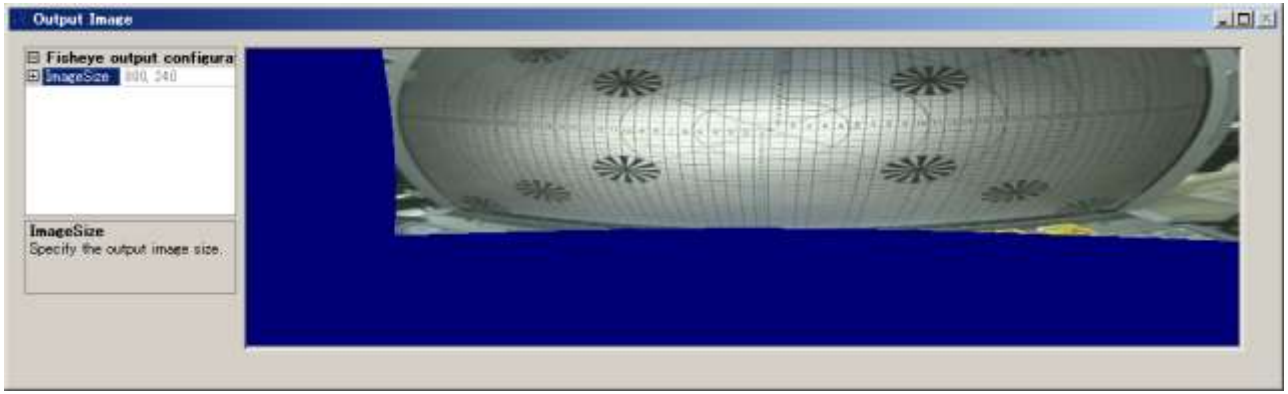
About FisheyeLite configuration. (9)

CenterPosition

CenterPosition = 400,120



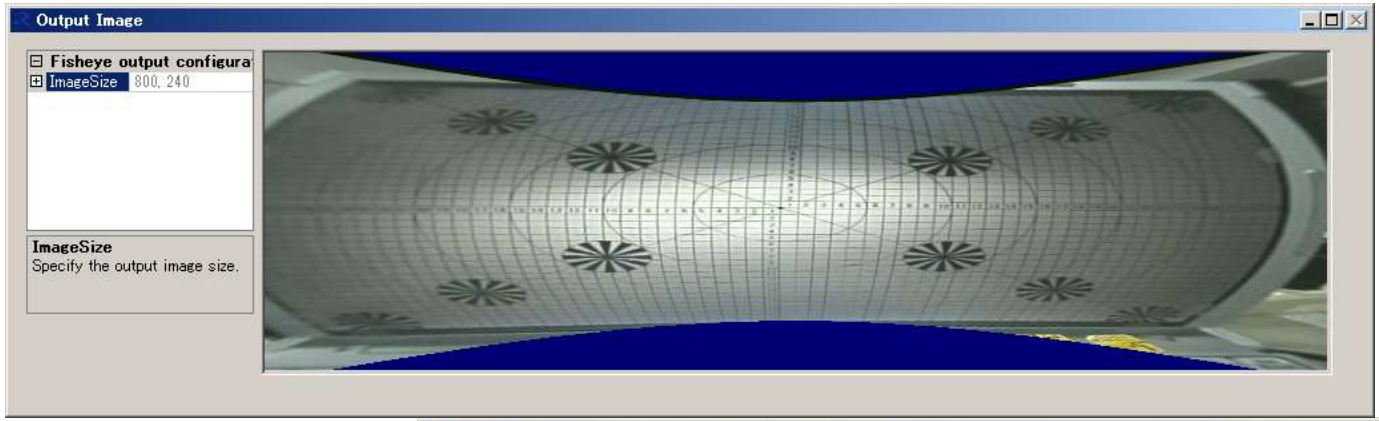
CenterPosition =350,200



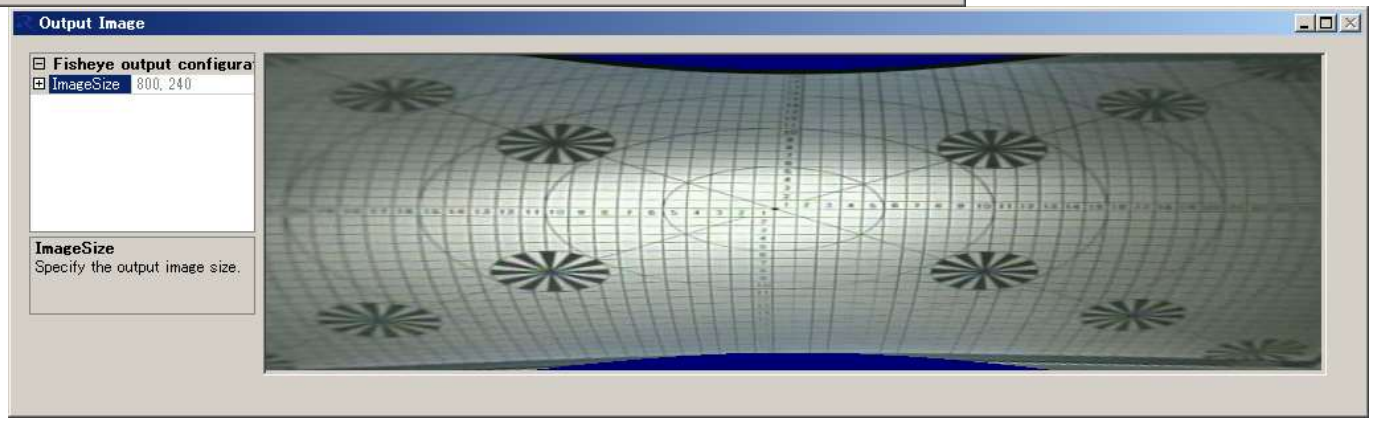
About FisheyeLite configuration. (10)

LensType

equidistant



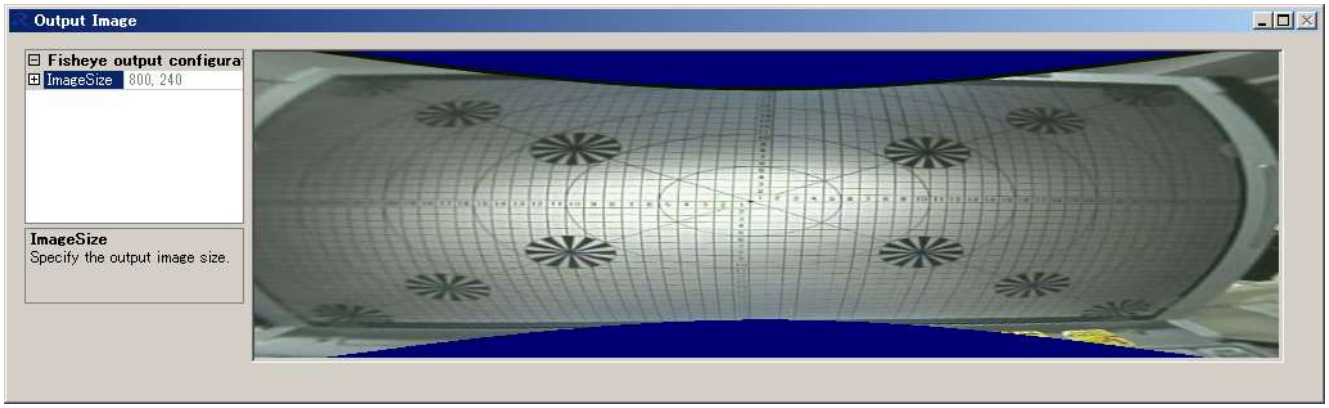
stereographic



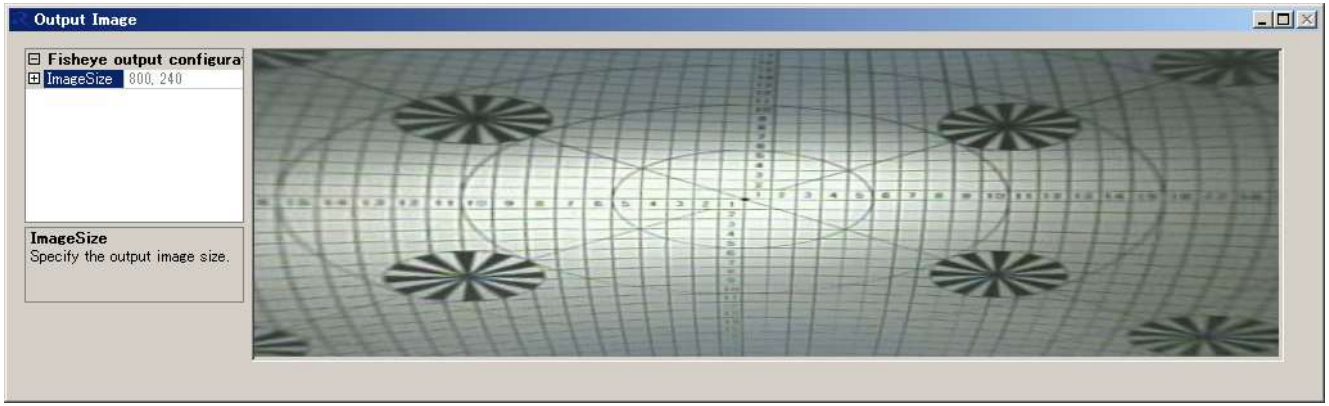
About FisheyeLite configuration. (11)

Radius

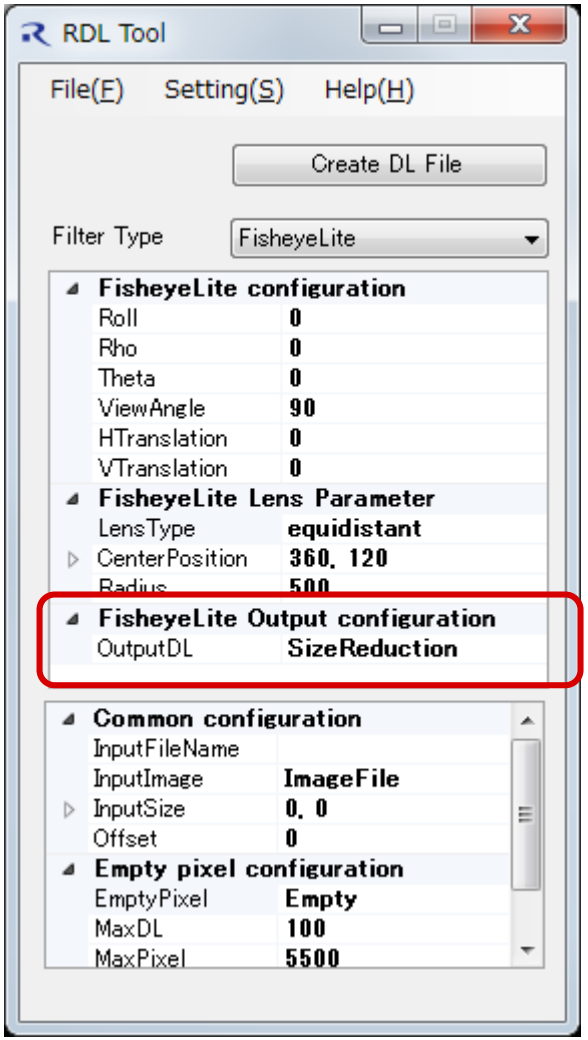
Radius = 700



Radius = 500



About FisheyeLite configuration. (12)



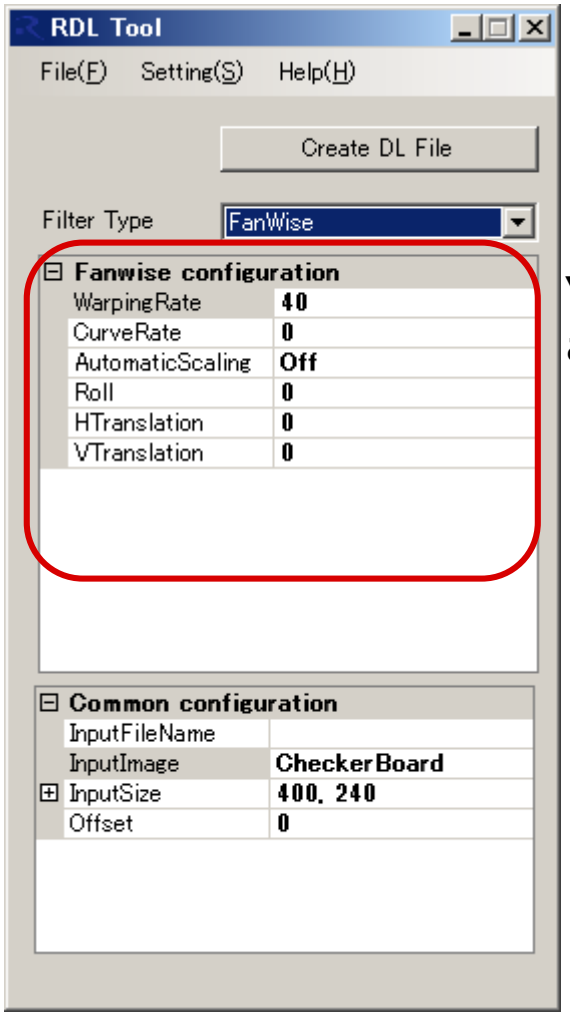
"FisheyeLite Output configuration"
will be displayed in this area when "FisheyeLite" is selected to "Filter Type".

About FisheyeLite configuration. (13)

OutputDL

- Specify the mode of output DL.
  - *SizeReduction*
    - Output DL size is smaller than *Normal* mode.
      The DL will work normally in RH850/D1Mx, but the simulation image in Output Image window will have a little error.
  - *Normal*
    - Output DL size is larger than *SizeReduction* mode.
      The simulation image will be normal.

About Fanwise configuration. (1)



"Fanwise configuration" Will be displayed in this area when "FanWise" is selected to "Filter Type".

About Fanwise configuration. (2)

WarpingRate (1 ~ 80)

- Specify the keystone warping factor. A value of 40 is the neutral setting. Values smaller than 40 (39 ~ 1) will shorten the length of the bottom edge of the image. Values bigger than 40 (41~80) will shorten the length of the top edge of the image.

CurveRate (0 ~ 20)

- Specify the sector warping factor. A value of 0 is neutral.

AutomaticScaling (On / Off)

- Specify automatic scaling for fan-shaped warping.

Roll (-180 ~ +180)

- Specifies the rotation angle.

HTranslation (-100 ~ +100)

- Specify the translation number of pixels in the horizontal direction.

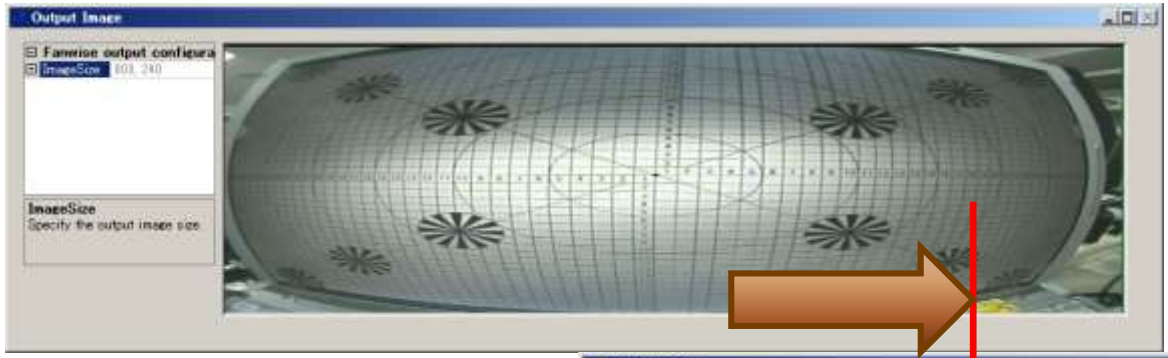
VTranslation (-100 ~ +100)

- Specify the translation number of pixels in the vertical direction.

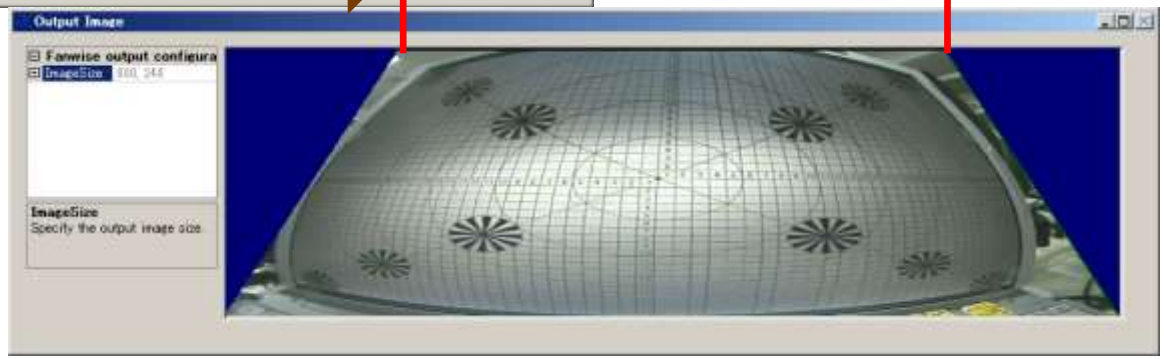
About Fanwise configuration. (3)

WarpingRate

WarpingRate = 40



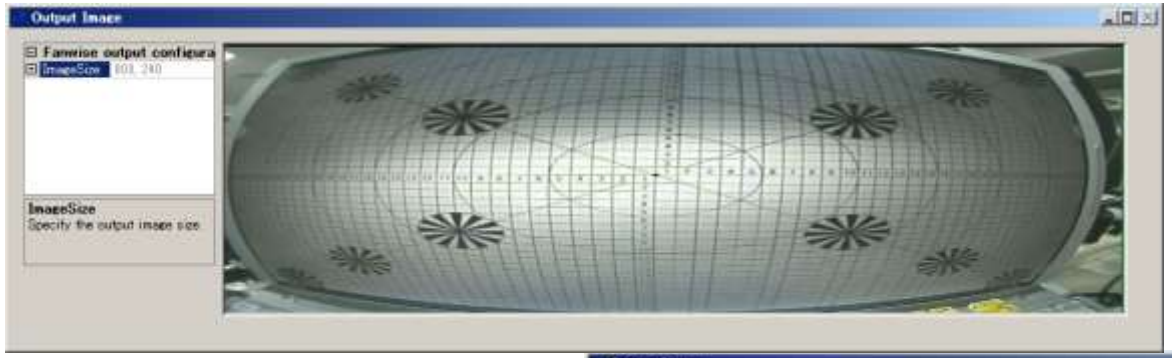
WarpingRate = 80



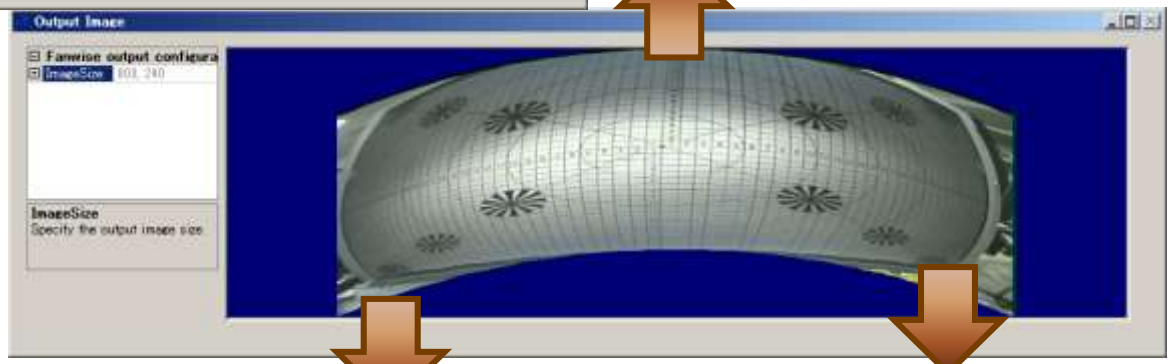
About Fanwise configuration. (4)

CurveRate

CurveRate = 0



CurveRate = 20

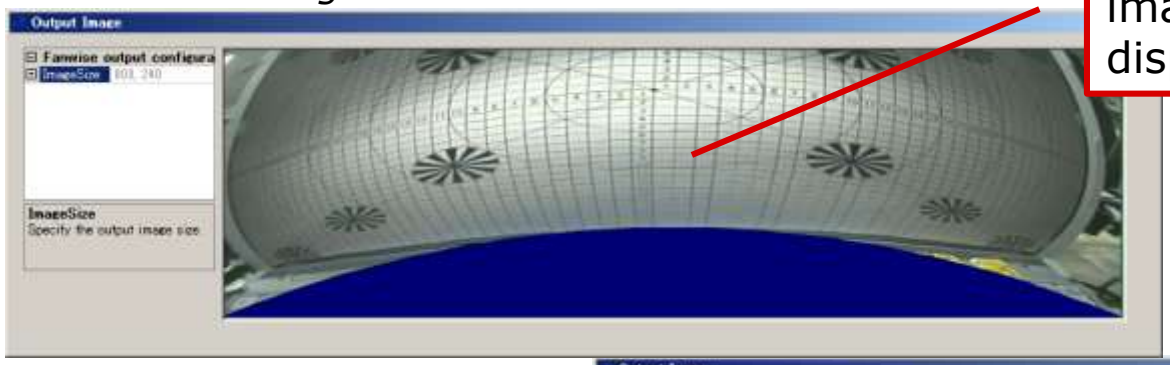


About Fanwise configuration. (5)

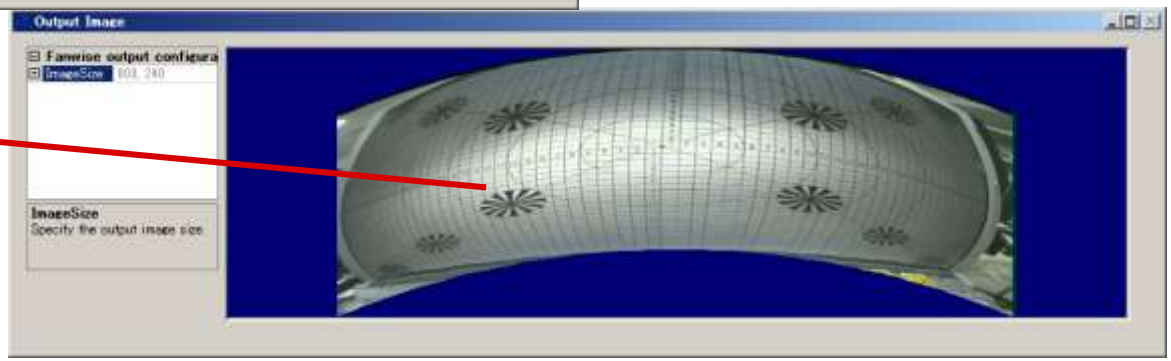
AutomaticScaling

Deformation may cause the output image to be displayed off screen.

AutomaticScaling = Off



AutomaticScaling = On

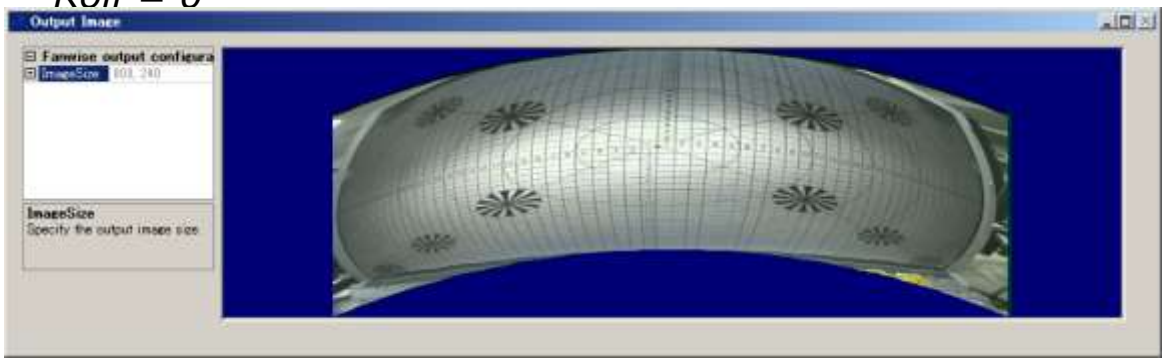


AutomaticScaling will reduce the image size to fit into the dimensions of the output image.

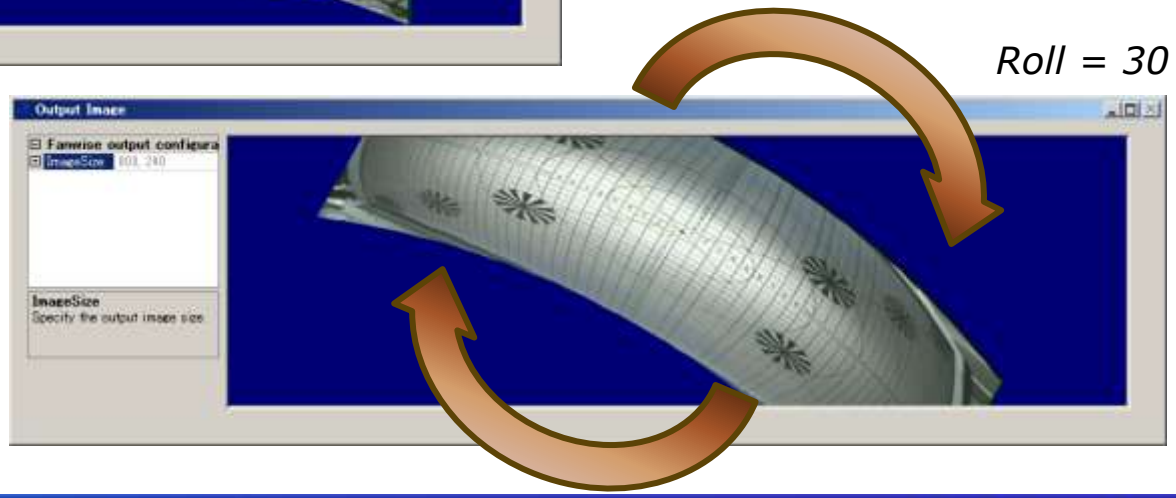
About Fanwise configuration. (6)

Roll

Roll = 0



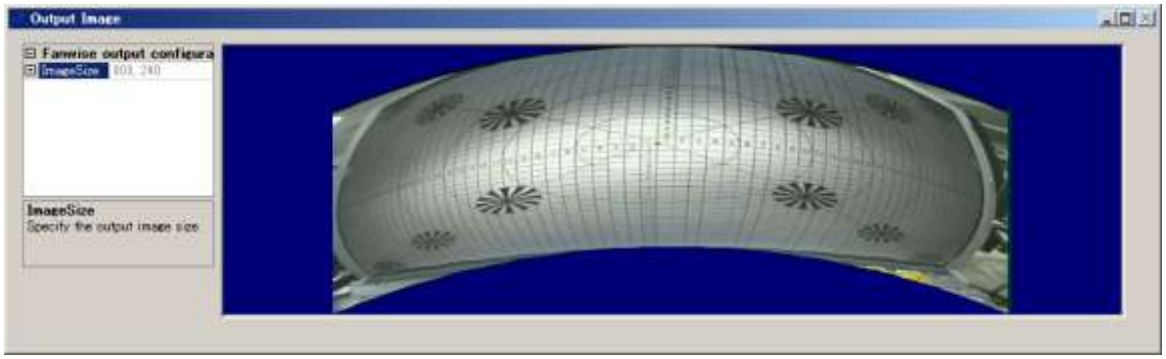
Roll = 30



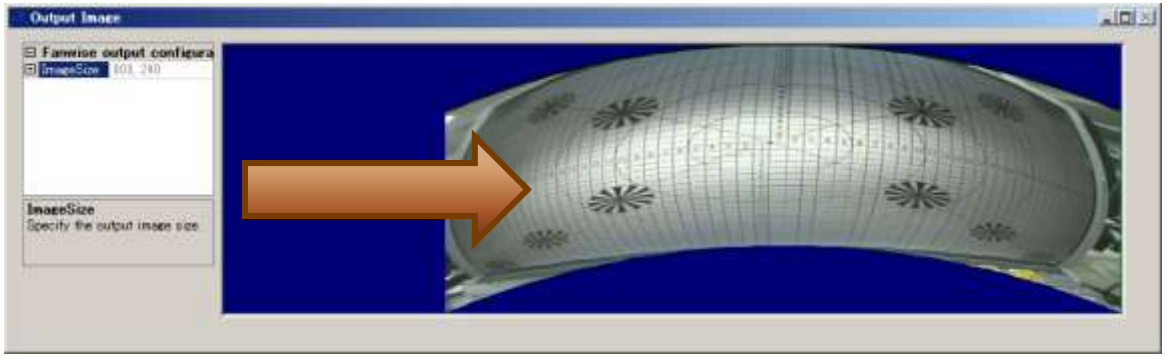
About Fanwise configuration. (7)

HTranslation

HTranslation = 0



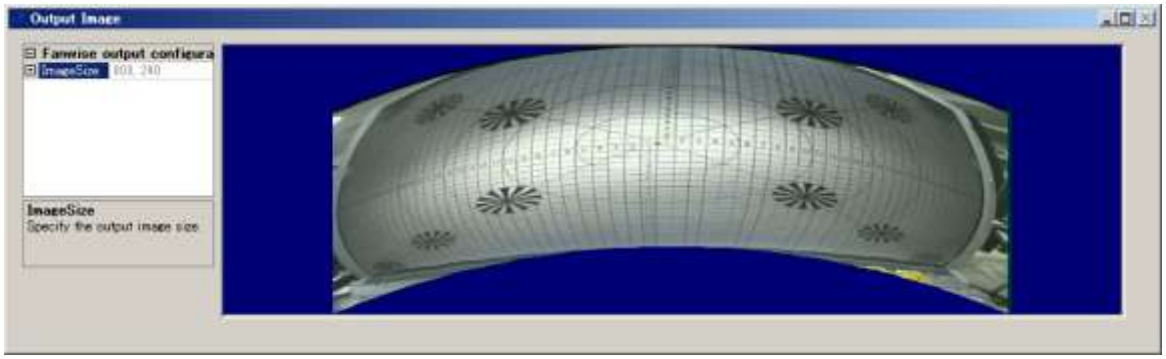
HTranslation = 100



About Fanwise configuration. (8)

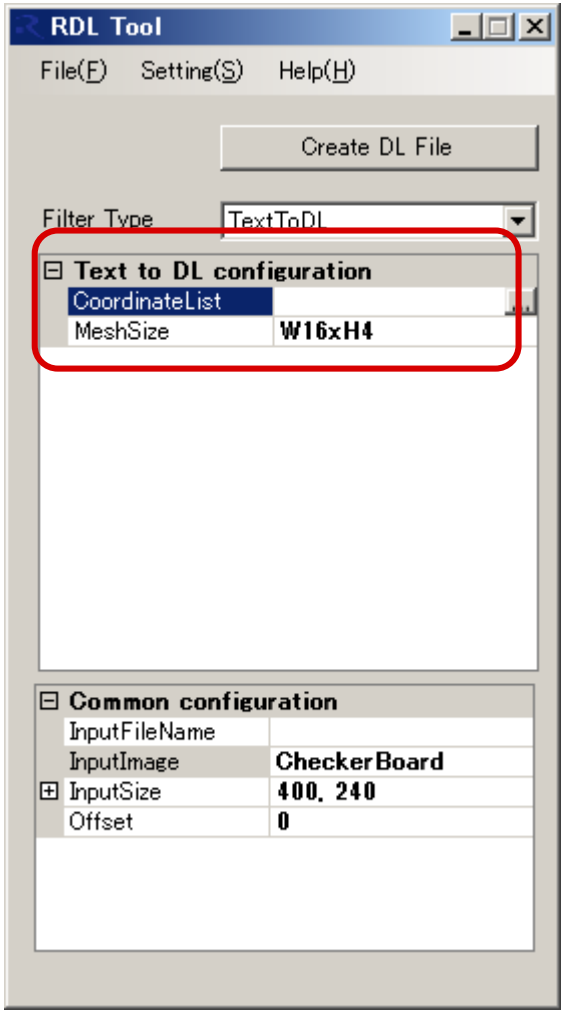
VTranslation

VTranslation = 0



VTranslation = 100

About Text file to DL configuration. (1)



"Text file to DL configuration" Will be displayed in this area when "Text file to DL" is selected to "Filter Type".

About Text file to DL configuration. (2)

CoordinateList

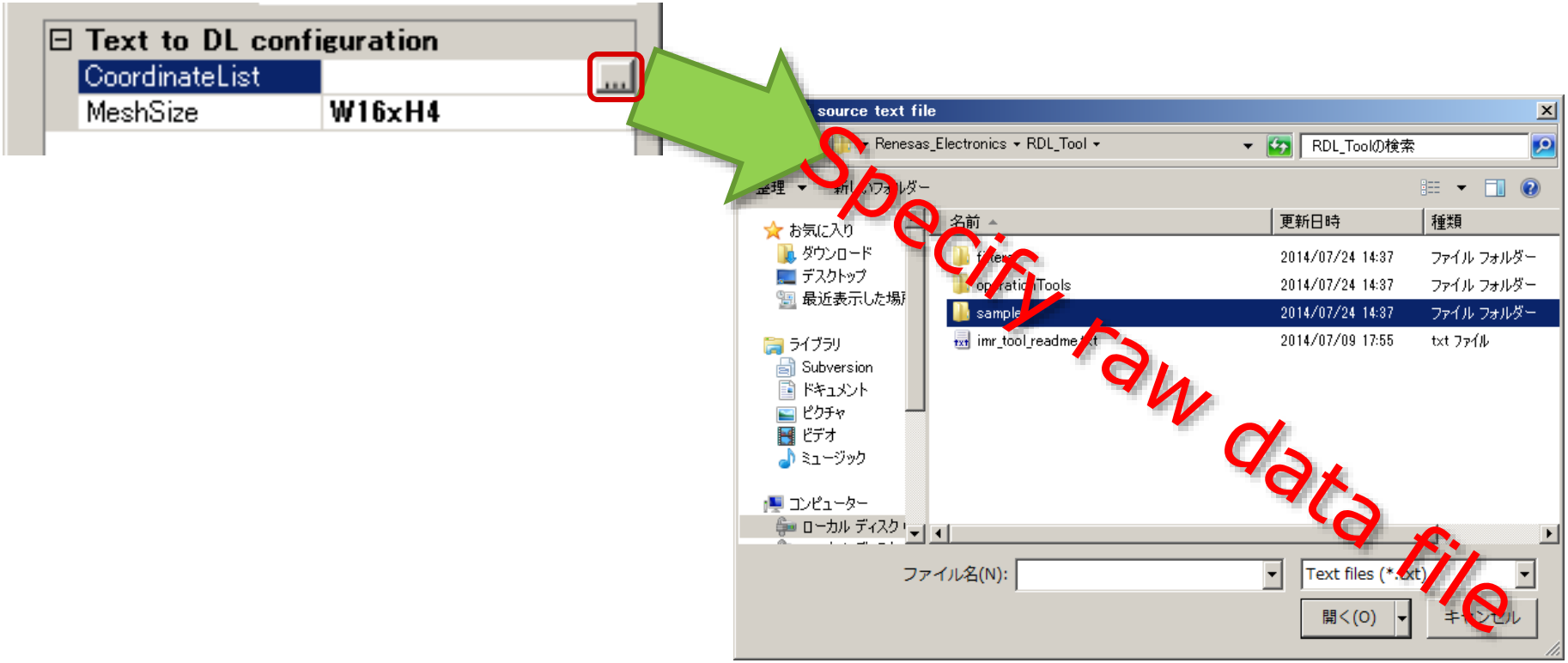
- Specify the filename of the text file containing the warping information.

Mesh Size

- Specify the mesh size of the warping information.

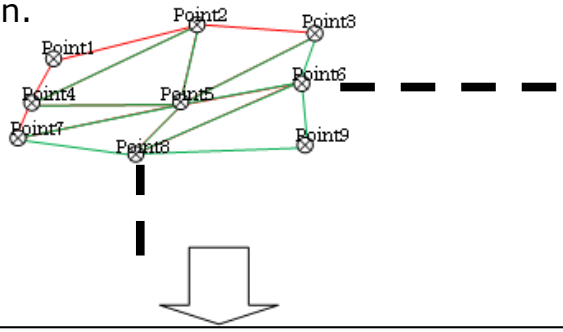
About Text file to DL configuration. (3)

Text Filename



About Text file to DL configuration. (4)

- The text file contains the destination coordinates for the warping operation. The source coordinates are automatically defined by the mesh size. For each source coordinate pair, the file must contain a destination coordinate pair. Each line of the file represents one line of laterally connected points of the mesh.
- It is an error if the number of points that are on one line is less than (1 + [ImageWidth] / [MeshWidth]). Unnecessary coordinate pairs are truncated and a warning will be displayed. The same rules apply to the vertical direction.



X of Point1, Y of Point1, X of Point2, Y of Point2, X of Point3, Y of Point3, …
X of Point4, Y of Point4, X of Point5, Y of Point5, X of Point6, Y of Point6, …
…

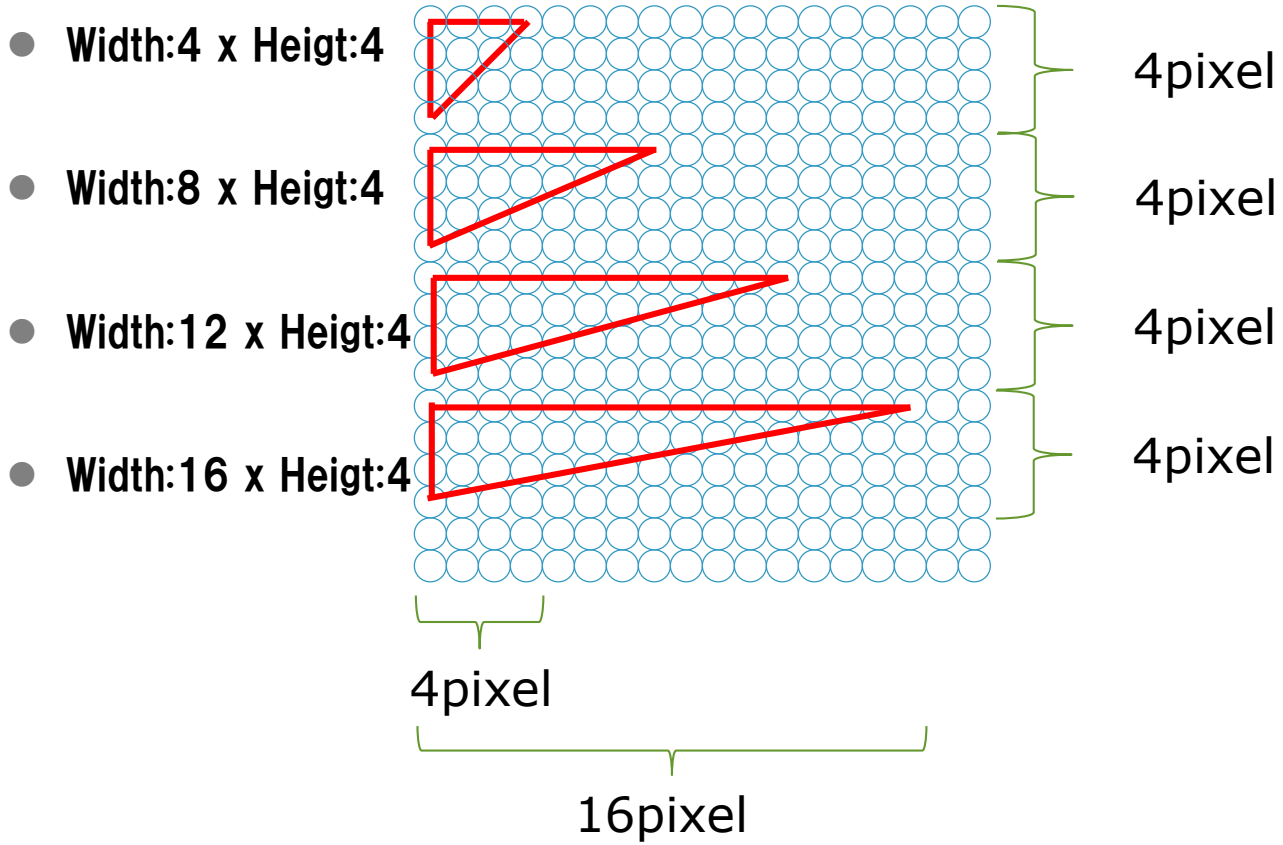
[Example of Text file for input]

```
0, 0, 16, 0, 32,0,  48,  0,64,  0,80,  0,96,  0,112,  0,128,  0,144,  0,160,  0,176,  0,192,  0,208,  0,224,  0,240,  0,
0, 4, 16, 4, 32,4,  48,  4,64,  4,80,  4,96,  4,112,  4,128,  4,144,  4,160,  4,176,  4,192,  4,208,  4,224,  4,240,  4,
0, 8, 16, 8, 32,8,  48,  8,64,  8,80,  8,96,  8,112,  8,128,  8,144,  8,160,  8,176,  8,192,  8,208,  8,224,  8,239,  8,
   (snip)
0,312,16,312,32,312,48,312,64,312,80,312,96,312,112,312,128,312,144,312,160,312,169,312,179,312,189,312,198,312,201,312,
0,316,16,316,32,316,48,316,64,316,80,316,96,316,112,316,128,316,144,316,160,316,169,316,179,316,189,316,198,316,201,316,
0,320,16,320,32,320,48,320,64,320,80,320,96,320,112,320,128,320,144,320,160,320,168,320,179,320,188,320,198,320,200,320,
```

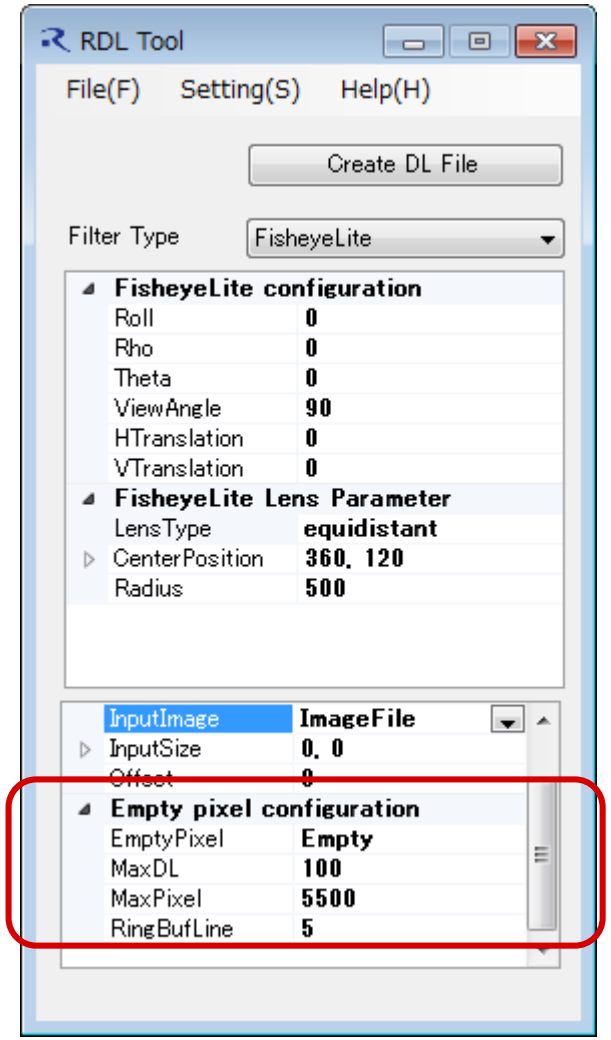
About Text file to DL configuration. (5)

Mesh Size

- **Choose the source mesh size**



About Empty pixel configuration. (1)



Depending on the filter parameters, some pixels may not be written.
In order to initialize these pixels, RDL tool supports to fill the empty pixels with black color.
Main usage of this feature is clearing the edge regions of a ring buffer. For non-ring buffer use cases, it is sufficient to clear the buffer once during initialization.

"Empty pixel configuration" will be displayed in this area at any time.

About Empty pixel configuration. (2)

Empty Pixel

- Specify the additional control over empty pixels from following types.
  - Empty (Empty pixels will not be written.)
  - FillSingleColor_normal (Fill with the single color with normal analysis.)
  - FillSingleColor_precision (Fill with the single color with precision analysis.)

Max DL

- Maximum number of data words (1 word is 32bit) to be processed per Mesh Line (4 Lines) including the instructions and both the pixels to be cleared and the pixel to be warped.

Max Pixel

- Maximum number of pixels to be written per Mesh Line (4 Lines) including both the pixels to be cleared and the pixel to be warped.

Ring Buf Line

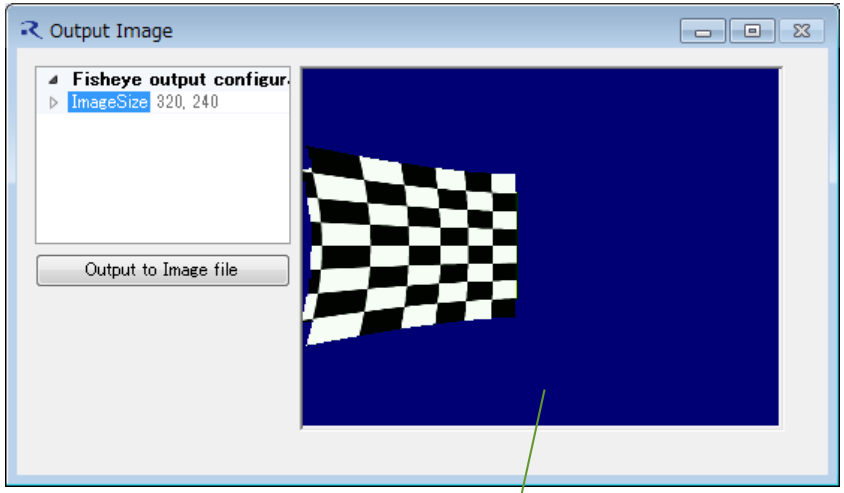
- Maximum distance between current input mesh line (source y-coordinate) and the mesh line being written (destination y-coordinate). In ring buffer mode, a high distance may cause visual artefacts due to wrapping of memory accesses. Reduce this value to prevent these effects.

"Max DL", "Max Pixel" and "Ring Buffer Line" are used to adjust the position of insertion of the clear instruction when "Empty Pixel" is set to FillSingleColor_normal or FillSingleColor_precision.
If RDL tool detects a error with insertion of clear instruction, it will turn back the output as "Empty Pixel = Empty".

About Empty pixel configuration. (3)

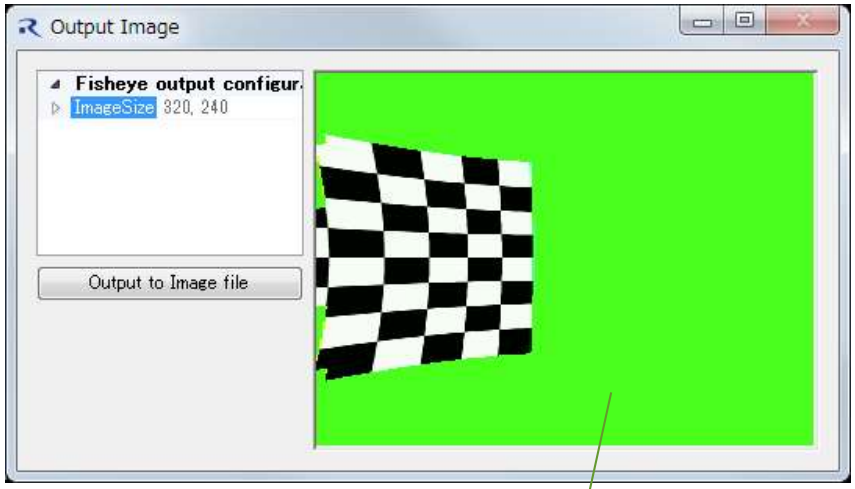
Empty Pixel

Empty Pixel = Empty



Blue area is not filled.

Empty Pixel = FillSingleColor_normal
or
Empty Pixel = FillSingleColor_precision

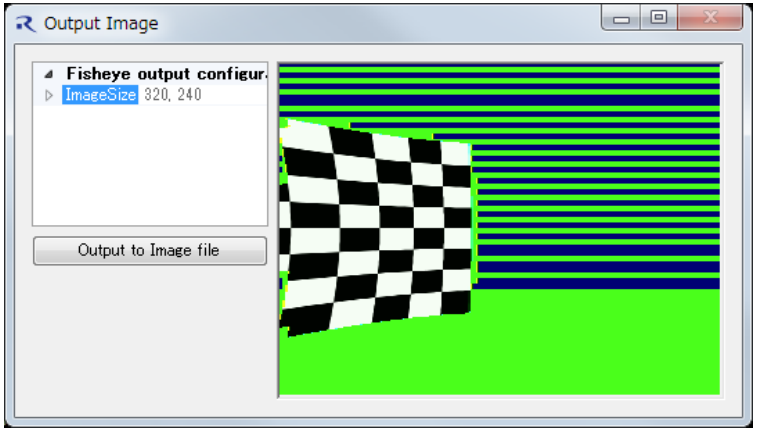


Green area is filled with single color.
(Single color is black.)

About Empty pixel configuration. (4)

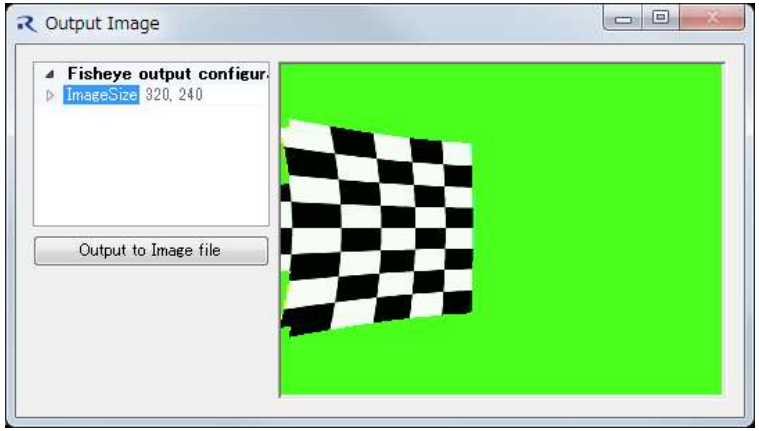
Max DL

Empty Pixel =FillSingleColor_precision
Max DL = 10
Max Pixel = 3000
Ring Buf Line = 0



If "Max DL" is too small, empty area is remained.

Empty Pixel =FillSingleColor_precision
Max DL = 200
Max Pixel = 3000
Ring Buf Line = 0

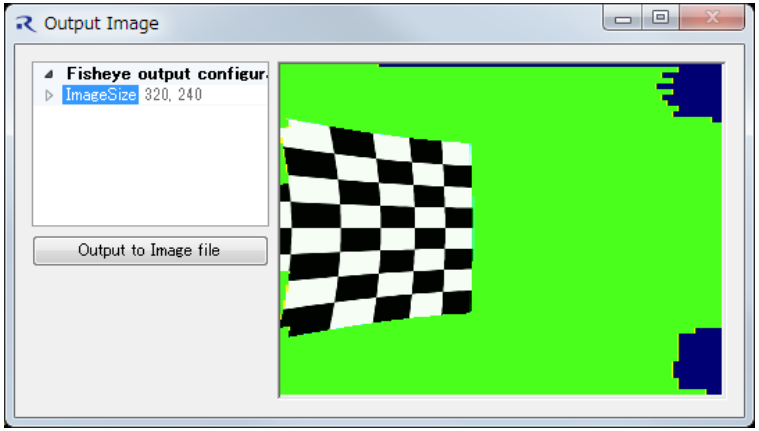


If "Max DL" is enough value, all area can be filled.
But, if "Max DL" is too large, it may not be processed normally on RH850/D1Mx.

About Empty pixel configuration. (5)

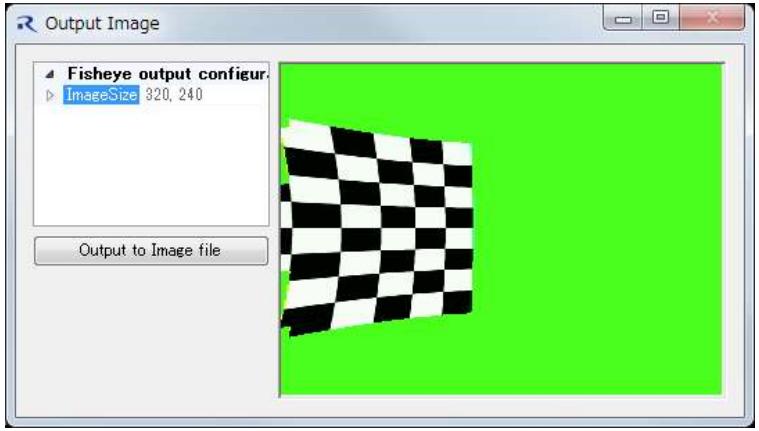
Max Pixel

Empty Pixel =FillSingleColor_precision
Max DL = 200
Max Pixel = 1500
Ring Buf Line = 0



If "Max Pixel" is too small, empty area is remained.

Empty Pixel =FillSingleColor_precision
Max DL = 200
Max Pixel = 3000
Ring Buf Line = 0



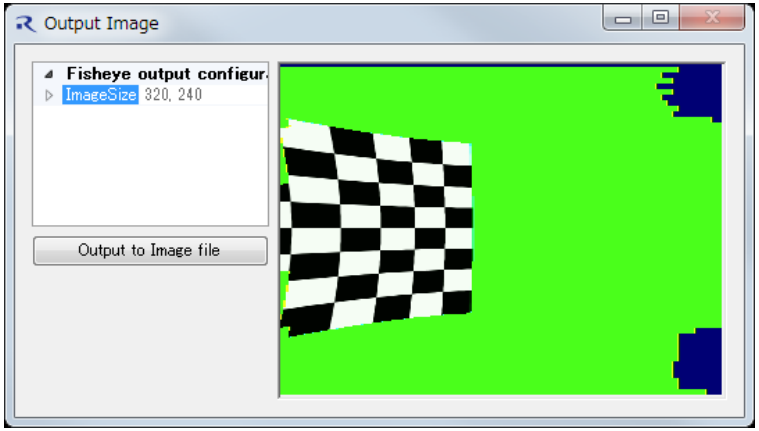
If "Max Pixel" is enough value, all area can be filled.
But, if "Max Pixel" is too large, it may not be processed normally on RH850/D1Mx.

"Max Pixel" should be smaller than following value at least.
"Max Pixel" < ((RH850 CPU frequency)[MHz] * 0.5 [pixel/clock] * (Hsync period)[usec] * 4 [line])
Please note that it does not mean to guarantee the maximum value.

About Empty pixel configuration. (6)

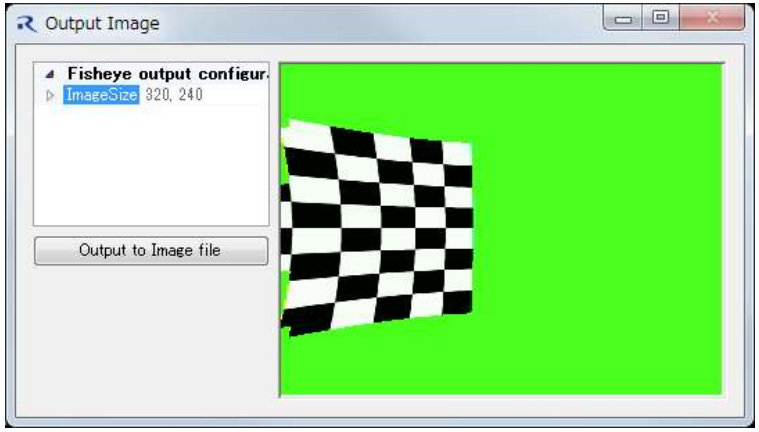
Ring Buf Line

Empty Pixel =FillSingleColor_precision
Max DL = 100
Max Pixel = 1500
Ring Buf Line = 0



If "Ring Buf Line" is too small, empty area is remained.

Empty Pixel =FillSingleColor_precision
Max DL = 100
Max Pixel = 1500
Ring Buf Line = 1



If "Ring Buf Line" is enough value, all area can be filled. But, if "Ring Buf Line" is too large, it may not be processed normally on RH850/D1Mx with ring buffer mode.

If you are using the ring buffer mode of the VOWE, "Ring Buf Line" should be smaller than holding lines in ring buffer. ("Ring Buf Line" < ((Ring buffer size)[Byte] / stride[Byte] / 4 [line]))
Please note that it does not mean to guarantee the maximum value.
If you are using the frame buffer mode of the VOWE, you can set any value to "Ring Buf Line".

Notes

Notes

- If the warping image does not appear in the output Image window, you may want to improve by setting again the source image file in SourceFileName.

Revision history

Revision	Date	Description	Summary
0.00	2014/10/24	First issued	--
0.10	2014/07/10	Add Text file to DL filter.	--
0.20	2014/07/22	Change application name from “DL Tool” to “RDL Tool”. Change how to Install/Uninstall.	--
0.30	2014/07/29	Replaced as those of the 1.7.1.1 version image.	--
0.31	2014/10/10	Update to Ver.1.7.2.0 the executable file.	--
0.40	2015/10/07	Add following parameters as Empty pixel configuration. - Empty Pixel - Max DL - Max Pixel - Ring Buffer Line	--
0.50	2015/12/24	Add “Output DL” parameters as FisheyeLite Output configuration.	--
0.60	2016/01/25	Fix descriptions.	--

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